MIL-T-49260(CR)

10 June 1982

MILITARY SPECIFICATION

TERMINAL, COMMUNICATIONS AN/UGC-74A(V)

This specification is approved for use by US Army Communications Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Terminal, Communications AN/UGC-74A(V), which is composed of Teleprinter TT-744/UGC-74A(V), Keyboard, TT-745/UGC-74A(V), Case, CY-7735/UGC-74A(V) and Cover, MX-10038/UGC-74A(V) (see 6.8).

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, USACECOM, ATTN: DRSEL-PED EC-2, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5815

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

UU-P-547	Paper, Teletypewriter, Roll

Military

MIL-P-116	Preservation-Packaging Methods of
MIL-S-901	Shock Tests, H.I. (High Impact), Shipboard Machinery, Equipment and Systems, Requirements for (Navy)
MIL-P-11268	Parts, Materials and Processes Used in Electronic Communication Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-T-49261	Terminal, Communications AN/UGC-74A(V), TEMPEST Requirements for (U)

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables Inspection by Attributes
MIL-STD-118-114 24 March 1976	Electrical Characteristics of Digital Interface Circuits
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard Requirements for Electronic Equipment
MIL-STD-461A Notice 4	Electromagnetic Interference Charactreristics, Requirements for Equipment
MIL-STD-462 Notice 3	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-471	Maintainability Demonstration
MIL-STD-781C	Reliability Test: Exponential Distribution
MIL-STD-810C	Environmental Test Methods
MIL-STD-1280	Keyboard Arrangements
MIL-STD-1474	Noise Limits for Army Material
MIL-STD-1562	List of Standard Microcircuits

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publication form a part of this specification to the extent specified herein.

DOCUMENTS

Electronics Command Specification

EL-CP0000-0031	Electronic Equipment Design Requirements for the Nuclear Radiation Environment

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DRAWINGS

SM-D-915600	Teletypewriter AN/UGC-74( ) (V)3
SM-D-972467	Mount Assembly
SPI-1G00173	Special Packaging Instructions

(Copies of specifications, standards, documents, and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish First Article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and of Drawing SM-D-915600 (including the requirements for parts, materials and processes thereon).

3.3 Parts, materials, and processes: General. In addition to the requirements of this specification, the requirements of MIL-P-11268 shall apply. Microelectronic devices shall be selected from MIL-STD-1562.

3.4 Finish. The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings.

3.5 Marking.

3.5.1 General. Marking shall conform to Specification MIL-M-13231.

3.5.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

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3.5.3 Serial numbers. The following items shall have serial numbers:

Teleprinter TT-744/UGC-74A(V)
Keyboard TT-745/UGC-74A(V)

3.6 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.7 Preconditioning. The equipment shall be capable of meeting the requirements herein, after subjection to the preconditioning of 4.6.1 and 4.6.2.

3.8 Functional requirements for Terminal, Communications AN/UGC-74A(V).

3.8.1 Paper usage. The printer shall feed either single-ply or multiple-copy (original and two copies) roll paper in accordance with UU-P-547 (roll).

3.8.2 Visibility of copy. Each character typed shall be visible after the ribbon drops without use of a space or line feed.

3.8.3 Copy illumination. With the POWER switch in the ON position, the copy lamp shall illuminate the paper. Illumination shall be adjustable by clockwise rotation of the ILLUM knob from the OFF position to the MAX position.

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3.8.4 ASCII/BAUDOT control. The setting of the control shall determine whether the teletypewriter shall transmit and receive either ASCII (USA Standard Code for Information Exchange) or BAUDOT Code (International Telegraph Alphabet American Variation (ITA-2)).

3.8.5 Printer functions.

3.8.5.1 Local printer functions. Local machine functions shall be performed whether or not the machine is also responding to line signals.

3.8.5.1.1 Line feed. Depressing the LINE FEED switch shall advance the paper continuously until the switch is released.

3.8.5.2 Coded BAUDOT printer functions.

3.8.5.2.1 Carriage return. This code shall return printing position to left margin and suppress printing and spacing during operation.

3.8.5.2.2 Line feed. This code shall advance paper one or two lines as determined by the parameter set in the line feed sub-command.

3.8.5.2.3 Letters. This code shall shift the printer to the printing letters condition.

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3.8.5.2.4 Figures. This code shall shift the printer to the printing figures condition.

3.8.5.2.5 FIG S/J switch. This switch shall be used to select whether a Figure S or Figure J code (only applicable in BAUDOT) shall activate the audible alarm as follows:

Switch Positions	Received Code	Action
S	Figure S Code	Alarm/(00101) entered in buffer
S	Figure J Code	Apostrophe entered in buffer
J	Figure J Code	Alarm/(01011) entered in buffer
J	Figure S Code	Apostrophe entered in buffer

3.8.5.3 Coded ASCII printer functions.

3.8.5.3.1 Keyboard/print codes. Tables I, II and III define the keyboard/print codes. Table I defines the Control Key Codes, Table II defines the Upper Case Key Codes and Table III defines the Lower Case Key Codes.

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TABLE I. Control key codes

RECEIVED OR KEYTOP SYMBOL	TRANSMITTED ASCII CODE 1/	BAUDOT SYMBOL	TRANSMITTED BAUDOT CODE 1/
NUL	0000000	SP	00000
SOH	0000001	NONE	NONE
STX	0000010	NONE	NONE
ETX	0000011	NONE	NONE
EOT	0000100	NONE	NONE
ENQ	0000101	NONE	NONE
ACK	0000110	NONE	NONE
BEL	0000111	FIG S/J	00101/01011
BS/FIGS	0001000	FIG	11011
HT(TAB)	0001001	SP	00100
LF	0001010	LF	00010
VT	0001011	NONE	NONE
FF	0001100	NONE	NONE
CR	0001101	CR	01000
SO	0001110	NONE	NONE
SI	0001111	NONE	NONE
DLE	0010000	NONE	NONE
DC1	0010001	NONE	NONE
DC2	0010010	NONE	NONE
DC3	0010011	NONE	NONE
DC4	0010100	NONE	NONE
NAK	0010101	NONE	NONE
SYN	0010110	NONE	NONE
ETB	0010111	NONE	NONE
CAN	0011000	NONE	NONE
EM	0011001	NONE	NONE
SUB	0011010	NONE	NONE
ESC	0011011	NONE	NONE
FS	0011100	NONE	NONE
GS	0011101	NONE	NONE
RS	0011110	NONE	NONE
US	0011111	NONE	NONE
SP	0100000	SP	00100

1/ The least significant bit (LSB) (b_1) of each code group is located in the far right position. In serial bit by bit transmission, the coded characters shall be transmitted low order (LSB) (b_1) first.

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TABLE 1. Control key codes - Continued

RECEIVED OR KEYTOP SYMBOL	PRINTED CHARACTER		
	ASCII	BAUDOT	
	PRINT	ENTRY PRINT	TRANSMISSION PRINT
NUL	SP	SP	SP
SOH	SP	◊	NONE
STX	SP	◊	NONE
ETX	SP	◊	NONE
EOT	SP	◊	NONE
ENQ	SP	◊	NONE
ACK	SP	◊	NONE
BEL	SP/ALARM	SP/ALARM/APOSTROPHE	SP/ALARM/APOSTROPHE
BS/FIGS	SP	^	^
HT	SP	SP	SP
LF	LINE FEED	LINE FEED	LINE FEED
VT	SP	◊	NONE
FF	SP	◊	NONE
CR	RETURN TO LEFT MARGIN	RETURN TO LEFT MARGIN	RETURN TO LEFT MARGIN
SO	SP	◊	NONE
SI	SP	◊	NONE
DLE	SP	◊	NONE
DC1	SP	◊	NONE
DC2	SP	◊	NONE
DC3	SP	◊	NONE
DC4	SP	◊	NONE
NAK	SP	◊	NONE
SYN	SP	◊	NONE
ETB	SP	◊	NONE
CAN	SP	◊	NONE
EM	SP	◊	NONE
SUB	SP	◊	NONE
ESC	SP	◊	NONE
FS	SP	◊	NONE
GS	SP	◊	NONE
RS	SP	◊	NONE
US	SP	◊	NONE
SP	SP	SP	SP

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TABLE II. Upper case key codes

RECEIVED OR KEYTOP SYMBOL	TRANSMITTED ASCII CODE 1/	BAUDOT SYMBOL	TRANSMITTED BAUDOT CODE 1/
!	0100001	FIGS !	01101
"	0100010	FIGS "	10001
#	0100011	FIGS #	10100
$	0100100	FIGS $	01001
%	0100101	NONE	NONE
&	0100110	FIGS &	11010
'	0100111	FIGS '	01011
(	0101000	FIGS (	01111
)	0101001	FIGS)	10010
*	0101010	FIGS *	NONE
+	0101011	NONE	NONE
,	0101100	FIGS ,	01100
-	0101101	FIGS -	00011
.	0101110	FIGS .	11100
/	0101111	FIGS /	11101
0	0110000	FIGS 0	10110
1	0110001	FIGS 1	10111
2	0110010	FIGS 2	10011
3	0110011	FIGS 3	00001
4	0110100	FIGS 4	01010
5	0110101	FIGS 5	10000
6	0110110	FIGS 6	10101
7	0110111	FIGS 7	00111
8	0111000	FIGS 8	00110
9	0111001	FIGS 9	11000
:	0111010	FIGS :	01110
;	0111011	FIGS ;	11110
<	0111100	NONE	NONE
=	0111101	NONE	NONE
>	0111110	NONE	NONE
?	0111111	FIGS ?	11001
@	1000000	NONE	NONE
A	1000001	LTRS A	00011
B	1000010	LTRS B	11001
C	1000011	LTRS C	01110
D	1000100	LTRS D	01001
E	1000101	LTRS E	00001
F	1000110	LTRS F	01101

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TABLE II. Upper case key codes - continued

RECEIVED OR KEYTOP SYMBOL	TRANSMITTED ASCII CODE 1/	BAUDOT SYMBOL	TRANSMITTED BAUDOT CODE 1/
G	1000111	LTRS G	11010
H	1001000	LTRS H	10100
I	1001001	LTRS I	00110
J	1001010	LTRS J	01011
K	1001011	LTRS K	01111
L	1001100	LTRS L	10010
M	1001101	LTRS M	11100
N	1001110	LTRS N	01100
O	1001111	LTRS O	11000
P	1010000	LTRS P	10110
Q	1010001	LTRS Q	10111
R	1010010	LTRS R	01010
S	1010011	LTRS S	00101
T	1010100	LTRS T	10000
U	1010101	LTRS U	00111
V	1010110	LTRS V	11110
W	1010111	LTRS W	10011
X	1011000	LTRS X	11101
Y	1011001	LTRS Y	10101
Z	1011010	LTRS Z	10001
[	1011011	NONE	NONE
\	1011100	NONE	NONE
]	1011101	NONE	NONE
^	1011110	NONE	NONE
_	1011111	NONE	NONE
`	1100000	NONE	NONE

1/ The LSB (b_1) of each code group is located in the far right position. In serial bit by bit transmission, the coded character shall be transmitted low order (LSB) (b_1) first.

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TABLE II. Upper case key codes - continued

RECEIVED OR KEYTOP SYMBOL	PRINTED CHARACTER		
	ASCII	BAUDOT	
	PRINT 2/	LOCAL PRINT 3/	TRANSMISSION PRINT
!	!	!	!
"	"	"	"
#	#	#	#
$	$	$	$
%	%	◊	NONE
&	&	&	&
'	'	'	'
(	(	(	(
)	)	)	)
*	*	*	*
+	+	◊	NONE
,	,	,	,
-	-	-	-
.	.	.	.
/	/	/	/
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9
:	:	:	:
;	;	;	;
<	<	◊	NONE

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TABLE II. Upper case key codes - continued

RECEIVED OR KEYTOP SYMBOL	PRINTED CHARACTER		
	ASCII	BAUDOT	
	PRINT 2/	LOCAL PRINT 3/	TRANSMISSION PRINT
=	=	◊	NONE
>	>	◊	NONE
?	?	?	?
@	@	◊	NONE
A	A	A	A
B	B	B	B
C	C	C	C
D	D	D	D
E	E	E	E
F	F	F	F
G	G	G	G
H	H	H	H
I	I	I	I
J	J	J	J
K	K	K	K
L	L	L	L
M	M	M	M
N	N	N	N
O	O	O	O
P	P	P	P
Q	Q	Q	Q
R	R	R	R
S	S	S	S
T	T	T	T
U	U	U	U
V	V	V	V
W	W	W	W
X	X	X	X
Y	Y	Y	Y
Z	Z	Z	Z

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TABLE II Upper case key codes - continued

RECEIVED OR KEYTOP SYMBOL	PRINTED CHARACTER		
	ASCII	BAUDOT	
	PRINT 2/	LOCAL PRINT 3/	TRANSMISSION PRINT
⌐	⌐	◊	NONE
\	\	◊	NONE
⌙	⌙	◊	NONE
^	^	◊	NONE
←	←	◊	NONE
		◊	NONE

2/ If an ASCII character with a parity error is received a ◊ symbol shall be printed locally and the space code shall be transmitted.

3/ In BAUDOT local printout, when a ◊ is printed, it is not put into memory, i.e., nothing is printed during transmission.

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TABLE III. Lower case key codes

RECEIVED OR KEYTOP SYMBOL	TRANSMITTED ASCII CODE 1/	BAUDOT SYMBOL	TRANSMITTED BAUDOT CODE 1/
a	1100001	LTR A	00011
b	1100010	LTR B	11001
c	1100011	LTR C	01110
d	1100100	LTR D	01001
e	1100101	LTR E	00001
f	1100110	LTR F	01101
g	1100111	LTR G	11010
h	1101000	LTR H	10100
i	1101001	LTR I	00110
j	1101010	LTR J	01011
k	1101011	LTR K	01111
l	1101100	LTR L	10010
m	1101101	LTR M	11100
n	1101110	LTR N	01100
o	1101111	LTR O	11000
p	1110000	LTR P	10110
q	1110001	LTR Q	10111
r	1110010	LTR R	01010
s	1110011	LTR S	00101
t	1110100	LTR T	10000
u	1110101	LTR U	00111
v	1110110	LTR V	11110
w	1110111	LTR W	10011
x	1111000	LTR X	11101
y	1111001	LTR Y	10101
z	1111010	LTR Z	10001

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TABLE III. Lower case key codes - continued

RECEIVED OR KEYTOP SYMBOL	TRANSMITTED ASCII CODE 1/	BAUDOT SYMBOL	TRANSMITTED BAUDOT CODE 1/
{	1111011	NONE	NONE
|	1111100	NONE	NONE
}	1111101	NONE	NONE
~	1111110	NONE	NONE
DEL/LTRS	1111111	LTRS	11111

1/ The LSB (b_1) of each code group is located in the far right position. In serial bit by bit transmission, the coded character shall be transmitted low order (LSB)(b_1) first.

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TABLE III. Lower case key codes - continued

RECEIVED OR KEYTOP SYMBOL	PRINTED CHARACTER		
	ASCII	BAUDOT	
	PRINT	LOCAL PRINT	TRANSMISSION PRINT
a	A	A	A
b	B	B	B
c	C	C	C
d	D	D	D
e	E	E	E
f	F	F	F
g	G	G	G
h	H	H	H
i	I	I	I
j	J	J	J
k	K	K	K
l	L	L	L
m	M	M	M
n	N	N	N
o	O	O	O
p	P	P	P
q	Q	Q	Q
r	R	R	R
s	S	S	S
t	T	T	T
u	U	U	U
v	V	V	V
w	W	W	W
x	X	X	X
y	Y	Y	Y
z	Z	Z	Z
{	[	◊	NONE
|	\	◊	NONE
}	]	◊	NONE
~	^	◊	NONE
DEL/LTRS	SP	———	———

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3.8.5.3.2 TAB. When this code is encountered, the print position shall be moved to the smallest set tab position which is greater than the current print position. The next character is printed in the position. If the present print position is greater than or equal to the largest tab position, the tab shall be replaced by a carriage return.

3.8.5.3.3 CR. This code shall return printing position to left margin and suppress printing and spacing.

3.8.5.3.4 LF. This code shall advance paper one or two lines as determined by the parameter set in the LINE FEED Sub-command.

3.8.5.3.5 Parity error. The PARITY lamp shall light when an ASCII character with a parity error is received and parity is not inhibited. The parity function shall be disabled when receiving BAUDOT characters. The PARITY RESET switch shall reset the PARITY lamp, and when held on, shall perform lamp tests. The PARITY switch shall be used to select odd or even parity or to inhibit the parity check. When the received message is printed, the parity error symbol (diamond) shall be printed in place of all characters that contained parity errors on reception.

3.8.5.4 Automatic carriage return. After the printer has reached the end of line position, and upon entry of a character which would normally cause the printer to move to the right (end of line +1), the printer shall automatically insert a carriage return and line feed.

3.8.5.4.1 End of line signal. The END OF LINE warning lamp shall be on when the print position is within 6 character positions of the end of the line.

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3.8.5.5 Alarm and indicators.

3.8.5.5.1 Audible alarm.

3.8.5.5.1.1 Audible alarm adjustment. The audible alarm level shall be adjustable from maximum to no alarm.

3.8.5.5.1.2 Audible alarm actuation. The audible alarm shall be actuated under the following conditions:

a. By Figure S/J Codes being received (BAUDOT mode) as follows:

Switch Position	Received Code	Action
S	Figure S	Alarm for 0.3 ± 0.1 second
J	Figure J	Alarm for 0.3 ± 0.1 second

b. A steady-state alarm shall be activated when an ASCII character with a parity error is received (PARITY lamp lights). The PARITY RESET switch shall turn the parity alarm condition off and extinguish the lamp.

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c. The alarm shall be actuated for 0.3 ± 0.1 second when a BEL code is received in the ASCII mode.

d. By LINE lamp conditions as follows:

Interface	Signal	Alarm
High-Level Digital	Open-Line Condition	Steady-State Alarm
Low Level Digital	Continuous Space	Steady-State Alarm
Conditioned Diphase	Continuous Space	Steady-State Alarm

e. A steady-state alarm shall be actuated when a message, with precedance of IMMEDIATE or higher, has been received. The alarm condition shall be turned off when the message is printed.

3.8.5.5.2 Power indicator. The power on indicator shall be the copy lamps, and they shall light when the POWER switch is activated if the ILLUM control is on and primary power is available.

3.8.5.5.2.1 Battery back up indicator. The BAT. BACK UP indicator shall be activated when the system is in the Battery Back Up State and the POWER switch is "ON". All other indicators shall automatically be turned off.

3.8.5.5.3 Paper low indicator, The PAPER LOW lamp shall light when the paper supply is reduced to a diameter of 1-7/16" ± 1/16". Printing shall be stopped and the print drum shall automatically be turned off when the paper supply is depleted.

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3.8.5.5.4 Line indicator. The LINE lamp shall be on while mark and space signals are being received, and shall be off while the line is idle. For the high level digital interface only, an open line condition shall cause the LINE lamp to flash on and off. For the low level digital interface only, a continuous space shall cause the lamp to flash on and off. For the conditioned diphase interface only, a continuous space shall cause the lamp to flash on and off. The audible alarm shall be on while the LINE lamp is flashing.

3.8.5.5.5 Received message indicator. The MSG RCVD Lamp shall be activated when a message has been received. If message memory has a message with precedence of IMMEDIATE or higher, the lamp shall flash until all the IMMEDIATE or higher precedent messages have been printed. For lower precedent messages, the lamp shall light in a steady state until all messages have been printed.

3.8.5.5.6 Transmit indicator, The XMIT Lamp shall light during message transmission.

3.8.5.5.7 Memory full warning indicator. The MEMORY FULL lamp shall be activated when 25% or less of memory is available for allocation.

3.8.5.6 Switches and controls.

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3.8.5.6.1 PARITY switch. The parity shall be active when the ASCII/BAUDOT switch is in the ASCII position, and shall be inactive in the BAUDOT position. When active, the EVEN setting shall cause ASCII characters with even parity to be transmitted, and shall cause received characters to be checked for even parity. Setting the switch to the ODD position shall cause ASCII characters with odd parity to be transmitted and shall cause received characters to be checked for odd parity. Setting the switch to the INHIBIT position shall cause ASCII characters with indeterminate parity to be transmitted. No parity check shall be made on received characters when the switch is set to the INHIBIT position.

3.8.5.6.2 STOP BITS switch. The STOP BITS switch shall provide for transmission of one, or two stop bits.

3.8.5.6.3 STATE switch. The AN/UGC-74A(V) shall operate in any of three switch selectable states. The first shall be a Receive ONLY (RO) State which shall operate in two different ways depending upon the presence of message memory. If the message memory is physically missing, the RO State shall print characters as they are received. If message memory is present, the RO State shall operate as a buffered printer. In the Keyboard Send/Receive (KSR) State, a line shall be transmitted as soon as it is composed. Also, if a character is received when in this state, the characters shall be automatically printed as they are received. In the Intelligent Communications Terminal (ICT) State, the AN/UGC-74A(V) shall have full-duplex capability and can be used to compose, edit, transmit, receive, print, and store messages.

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3.8.5.6.4 Data inversion. The capability to invert low level transmitted and/or received data shall be provided.

3.8.5.6.5 Self test. The SELF TEST switch shall activate the Self Test State.

3.8.5.6.6 ABORT switch. Activation of the momentary ABORT switch shall abort all current processing and return control to the operator.

3.8.5.6.7 Audio alarm reset. The AUDIO ALARM RESET switch shall reset all current steady-state audible alarm conditions.

3.8.5.6.8 External power switch. The external power switch located on the front panel shall apply or disconnect primary power to the AN/UGC-74A(V).

3.8.6 System states.

3.8.6.1 Off state. This non-operational state shall be entered whenever the POWER switch is off and the system is powered down. This state shall also be entered whenever prime power is lost and no battery backup is available. In this state, all other states and processes shall be inoperative. Prior to starting up the system, manual switch settings shall be made in this state.

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3.8.6.2 Cold start state. When power is applied, this non-operational state shall be entered. The state of the system and memory shall be initialized to no messages present in message memory and the following message shall be printed.

"SYSTEM INITIALIZED"

The transfer to the next state (see 3.8.6.3) shall be automatic when initialization is complete.

3.8.6.3 Operation validation/state determination state. Upon entry into this non-operational state, the operation validation test shall be performed without operator intervention. This test shall determine if the system is operational without destroying the state of the machine. The system shall sample the internal switches and configurations. Based on this information, the state of the system shall be initialized and stored. The operational state of the system shall be determined by the following rules:

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STATE Switch Position	Message Operational Memory Present	Keyboard Present	State
ICT	YES	YES	ICT
ICT	NO	YES	KSR
ICT	NO	NO	RO
KSR	N.A.	YES	KSR
KSR	N.A.	NO	RO
RO	YES	N.A.	RO
RO	NO	N.A.	RO

The system shall print out the following information:

SWITCH STATE = ICT, KSR or RO

OPERATIONAL STATE = ICT, KSR or RO

OPERATING CAPACITY = FULL or DEGRADED

MODE = ASCII or BAUDOT

BAUD RATE = 45.5, 50, 75, 150, 300, 600 or 1200

STOP BITS = 1 or 2

END OF LINE OPTION = 0 to 8 ASCII characters representing END OF LINE sequence

SPACE OPTION = ON or OFF

LINE LENGTH = 40 to 80

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LINE FEEDS = 1 or 2

RECEIVE ENVELOPE OPTION = START RECEIVE ENVELOPE SEQUENCE AND STOP RECEIVE ENVELOPE SEQUENCE

TRANSMIT ENVELOPE OPTION = START TRANSMIT ENVELOPE SEQUENCE AND STOP TRANSMIT ENVELOPE SEQUENCE

In the BAUDOT mode, the following information shall be printed:

BELL OPTION = FIGURES S or FIGURES J

In the ASCII mode, the following information shall be printed:

PARITY OPTION = ODD, EVEN or INHIBIT

CAPITAL LETTER OPTION = ON or OFF

If any combination of switches produces an illegal configuration, an error message shall be printed. The system shall stop until power is turned off, the switches properly set and power is turned on again.

Invalid options shall be as follows:

a. ASCII at less then 75 baud

b. BAUDOT at greater than 75 baud

c. Conditioned diphase at less than 300 BAUD or in BAUDOT mode

d. Conditioned diphase with external clock

When the processing in this state is complete, the control shall be transferred to the appropriate state based on the state selected. The receive process shall be initialized and started.

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3.8.6.4 ICT state. In this operational state, the operator shall be able to control system operation via keyboard command entry. The system commands shall specify the functions available on the system. The Edit Mode shall allow the operator to create and change messages. The Parameter Mode shall allow system parameters and options to be specified. If power loss is detected, the system shall go to the Battery Back Up State. If the SELF TEST switch is activated, the system shall go to Self Test State.

3.8.6.5 RO State. In this operational state, if message memory is physically missing, all data shall be printed as received. If message memory is present, the machine shall operate as a buffered printer. Data shall be printed as it is received. In the event of receiving data faster than the machine can print, this data shall be stored in message memory (up to the capacity of the memory). The machine shall continue printing until all received data has been printed. If the printer receives data such that it overflows the buffer, that data shall be lost. No indication shall be given to the operator that data was lost. The RO State shall not interpret communications envelopes, end of line sequences, or redundant carriage returns.

3.8.6.6 KSR State. In this operational state, the basic editing functions (DLC, DLL and REV keys) shall be available. As each line is composed and a CR is entered, it shall be transmitted. During message transmission operator input shall not be accepted. Received characters shall be printed as they are received. In the event of receiving data faster than the machine can print, no data shall be lost. The KSR State shall not transmit or interpret communication envelopes, end-of-line sequences or redundant carriage returns.

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3.8.6.7 Battery Back Up State. When this state is entered, all data in message memory shall be retained and all current processing shall be terminated. Any composed data being used for current processing may be lost. If no back up battery power is provided, the system shall revert to the OFF State. If back up battery power is provided, the system shall turn on the BAT. BACK UP indicator and go into a non-operational state of minimum power consumption. When the system is in Battery Back Up State, the BAT. BACK UP indicator shall be the only indicator active. All other indicators shall be off. When prime power is returned, the system shall go to the Operation Validation/State Determination State.

3.8.6.8 Self Test State. The Self Test State shall be entered unconditionally via depression of the SELF TEST switch, except in Battery Back Up State. In the Self Test State, a series of tests shall be performed that indicate the system status as operational or non-operational. The operator may participate in the testing. Upon successful completion of the tests, the word "READY" shall be printed, followed by a carriage return with a line feed. Upon failure, faults shall be isolated to the optimum replaceable subassembly.

3.8.7 Message processing. A line edit input processing capability shall be provided. This capability shall permit three basic editing functions. The functions shall be:

a. Delete last character entered (DLC key)

b. Delete current input line (DLL key)

c. Reprint current input line (REV key)

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These basic edit keys shall be active at all times for all keyboard inputs during the current line. The current line shall be defined as any key-strokes up to but not including the carriage return or 80 characters, whichever comes first.

3.8.7.1 Message storage. Messages shall be stored up to the maximum of memory capacity.

3.8.7.1.1 Memory full.

3.8.7.1.1.1 Operator input. When message memory available becomes 384 characters or less due to an operator input, the line entered by the operator shall not enter message storage, and a special message shall be printed. To alleviate the message memory full condition, the operator shall be able to clear memory by deleting messages via a processing command.

3.8.7.1.1.2 Incoming messages. When message memory available becomes 384 characters or less due to an incoming message, the operator shall be pre-empted, and the printer taken over by the system (keyboard shall be locked out). The current line (data or command) shall be lost and a special message shall be printed, followed by 10 line feeds. The system shall respond in one of the two following ways depending upon the presence or absence of a received message in memory. When there are received messages in memory, they shall be printed out and subsequently deleted from memory in order of oldest-highest precedence first. These messages shall be separated by 10 line feeds. As memory blocks of data have been printed, the incoming message shall use these memory blocks for storage. When there are no received messages in message storage, the message being received shall be printed from the beginning. As memory blocks of data have been printed, the incoming message shall use these memory blocks for message storage. This process shall continue until all received messages are printed and deleted from message storage.

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This process is only applicable in the ICT Operational State. The machine shall return control to the operator in the system command level.

3.8.7.2 Message receipt. The AN/UGC-74A(V) shall be able to receive messages at all times that prime power is on. This shall be an automatic non-operator selectable function. The equipment shall recognize an end of message sequence, only in the ICT Operational State. Received messages shall be assigned a unique local message number between 1 and 9999. The MSG RCVD lamp shall be activated when the message is marked as received.

3.8.7.2.1 Receive Message Process. The Receive Process shall consist of the following sequence of events:

a. If the front of receive envelope is not null, then the incoming characters shall be compared and discarded until the envelope sequence is recognized. The next character shall be considered the first of the message. If the front of receive envelope sequence is null, the first character received shall be considered first character in the message.

b. If the system is in ASCII mode, the characters shall be saved in memory. If a parity error is detected and the PARITY switch is enabled, the PARITY lamp shall be turned on and a diamond code is stored. If the system is in BAUDOT mode, the characters shall be saved in memory. Upon message receipt, the end of line sequence shall be automatically replaced with a single CR.

c. If the close of receive envelope sequence is not null and no characters are received for 10 seconds, the message shall be assumed lost. The memory used so far shall be released and the receive process shall go back to looking for another message.

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If the close of receive envelope character sequence is detected, the message shall be considered as complete and the information shall be provided to the system so it may become a received message. The Receive Process shall return to looking for another mesesage.

d. If the close of receive envelope sequence is null and no characters are received for 10 seconds, the message shall be considered as complete. If the front of receive envelope sequence is not null and this sequence detected again, the message shall be considered as complete. The information shall be provided to the system so it may become a received message and the Receive Process shall return to looking for another message.

e. If the front of receive envelope is null or defined and found, the message shall be considered open and active until terminated as described below:

FRONT ENVELOPE		CLOSE ENVELOPE		TIME-OUT 10 SEC.	NEXT FRONT ENVELOPE FOUND	ABORT SWITCH	CURRENT MESSAGE STATUS
NULL	DEFINED and FOUND	DEFINED	FOUND				
X				X			SAVE
X		X		X			DUMP
X		X	X				SAVE
X						X	DUMP
	X			X			SAVE
	X				X		SAVE
	X	X		X			DUMP
	X	X			X		DUMP
	X	X	X				SAVE
	X					X	DUMP

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f. The Receive Process shall be terminated by the system if the ABORT switch is activated. In this case, the message currently being received (if any) shall be discarded, the memory used so far shall be released, and Receive Process shall return to looking for another message.

3.8.7.3 Transmit Message Process. The transmit Message Process shall be operator initiated through a system command. The Transmit Message Process shall consist of first sending out the front of transmit envelope sequence. The message shall be accessed from memory and the characters transmitted sequentially in BAUDOT or ASCII code. In serial bit by bit transmission, the coded character shall be transmitted low order, least significant bit (LSB)(b_1) first. When a CR is encountered in the message, it shall be replaced by the end of line sequence. The close of transmit envelope shall be appended to the message. The system shall now recognize the message as having been transmitted and the process terminated. The transmit process can be terminated before completion by the system if the ABORT switch is activated. The transmitted message shall be automatically printed, unless the halt key is activated. The halt key shall terminate printing, but not transmission.

3.8.7.4 System commands. All commands shall be leading and trailing space insensitive and truncated forms of commands shall be allowed. Either capital or lower case characters shall be accepted. The system shall prompt with a System Command prompt sequence.

3.8.7.4.1 PARAMETER Command. The system shall enter the PARAMETER Mode and prompt with the PARAMETER prompt sequence. PARAMETER Sub-commands shall be the only valid commands or sub-commands until the EXIT Sub-command is executed.

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3.8.7.4.1.1 LINE Sub-command. This sub-command shall allow the operator to specify the length of the text line for message composition and editing. If no value is specified, then 80 shall be used. The value specified shall be between 40 and 80 inclusive or an error message shall be printed and sub-command completed. On power-up initialization, the line length shall be set at 80.

3.8.7.4.1.2 SETAB Sub-command. This sub-command shall allow the operator to specify print positions at which tabs shall be set. This sub-command shall take a list of one or more tab parameters. If any of the tab parameters is greater than the current line length, an error message shall be printed and only the tab parameters that precede the invalid entry in the sub-command shall be changed. If a specified value is already a set tab, then no change shall be made. If the combination of previously set and currently being set tabs results in more than 12 tabs, the entries preceedig the entry which causes the setting of the thirteenth tab shall be accepted, then the sub-command shall be considered completed and an error message shall be printed. The tabs shall be displayed after the sub-command.

3.8.7.4.1.3 CLRTAB Sub-command. This sub-command shall allow the operator to clear tabs in the system. If "*" is specified, then all tabs shall be removed. On power-up initialization all tabs shall be cleared. This sub-command shall take a list of one or more parameters. If a specified tab value is not set, no action shall be taken. The remaining tabs set shall be displayed after this command. If none are set, an error message shall be printed. Only the CLRTAB entries preceding an invalid entry shall be accepted.

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3.8.7.4.1.4 DISPLAY Sub-command. All tab settings of the system shall be displayed in ascending order. If none are set, a message shall be printed.

3.8.7.4.1.5 CAPIT Sub-command. This sub-command (ASCII mode only) shall allow the operator to set or reset the capitalize option. This option shall be always set ON in the BAUDOT mode. If this option is set, all lower case letters (a through z) input from the keyboard shall be converted to capitals and stored as such in memory. If this option is not set, no change shall be made. On power-up initialization this option shall be active.

3.8.7.4.1.6 REMIND Sub-command. This sub-command shall allow the operator to set or reset the remind option. If the option is set, the following reminder shall be printed whenever a TRANSMIT or TTY (in ICT mode) Command is given:

KEY TRANSMITTER AND ENTER CR

On power-up initialization this option shall be inactive.

3.8.7.4.1.7 EOL Sub-command. This sub-command shall allow the operator to set the end of line (EOL) sequence for transmission. During transmission of a message, when a CR is found in the buffer, the EOL character sequence shall be sent in place of the CR. The EOL character sequence shall consists of zero to eight characters. If more than eight characters are specified or if the EOL sequence is improperly specified, an error message shall be printed. On power-up initialization, the EOL sequence shall be:

Characters: CR CR LF

If transmission is in process when the EOL Sub-command is recognized by the system, an error message shall be printed, no change shall be made and the sub-command shall be complete.

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3.8.7.4.1.8 LF Sub-command. This sub-command shall allow the operator to specify either one or two line feeds following a CR when printing. The operator must specify either 1 or 2 or an error message shall be printed and the value shall not be changed. On power-up initialization, line feed shall be set at 1.

3.8.7.4.1.9 SPACE Sub-command. This sub-command (ASCII mode only) shall allow the operator to set the spaces option. This option shall be always set ON in the BAUDOT mode. When the "SPACE" option is either OFF or ON, and there are no more tabs between the present print position and the EOL position, a CR (carriage return) shall be inserted into memory whenever a tab character is recognized as input from the keyboard. When the "SPACE" option is ON, and there is a tab between the present print position and the EOL position, space codes shall be inserted into memory (up to but not including the tab position) whenever a tab character is recognized as input from the keyboard. When the "SPACE" option is OFF, and there is a tab between the present print position and the EOL position, the tab code shall be inserted into memory whenever a tab character is recognized as input from the keyboard. If "ON" or "OFF" is not inserted in the parameter field, an error message shall be printed and no change made. On power-up initialization, the space option shall be OFF when in the ASCII mode.

3.8.7.4.1.10 PAPER Sub-command. This sub-command shall turn off the print drum and ribbon feed so that the paper and/or ribbon may be changed without loss of memory integrity. No further inputs shall be printed until the motor is restarted. Any keys may be activated but they shall be ignored until the specific sequence GO (CR) is entered. The LINE FEED switch shall be active to assist in loading the paper. After GO (CR) is entered, the system shall respond by turning the motor back on and returning to the previous command or sub-command level. Messages shall be lost if the memory becomes full while the printer is inactive. If the paper runs out, the system shall automatically initiate this command and stop the print drum.

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3.8.7.4.1.11 RENVELOPE Sub-command. This sub-command shall allow the operator to set the envolope which shall be removed from a message on reception by the system. This sequence shall never appear in message storage and shall be used for communications purposes only. The envelope shall consist of zero to eight characters of start sequence and zero to sixteen characters of stop sequence. If more than the required characters are specified, or improperly specified, an error message shall be printed. If a message is currently being received when the RENVELOPE Sub-command is recognized by the system, an error message shall be printed, no change shall be made, and the sub-command shall be complete. On power-up initialization, if the system is in BAUDOT mode the envelope shall be null. If the system is in ASCII mode the envelope shall be:

	Front	END
ASCII Characters:	V Z C Z C :	N N N N

3.8.7.4.1.12 TENVELOPE Sub-command. This sub-command shall allow the operator to set the envelope which shall be put around a message for transmission by the system. This sequence shall never appear in the message storage and shall be used for communications purposes only. The envelope shall consist of zero to eight characters of start sequence and zero to sixteen characters of stop sequence. If more than the required characters are specified, or improperly specified, an error message shall be printed. If a message is currently being transmitted when the TENVELOPE Sub-command is recognized by the system, an error message shall be printed, no change shall be made, and the sub-command shall be complete. On power-up initialization, if the system is in BAUDOT mode the envelope shall be null: If the system is in ASCII mode the envelope shall be:

	Front	END	12 x
ASCII Characters:	V Z C Z C :	N N N N	DEL - - DEL

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3.8.7.4.1.13 EXIT Sub-command. This sub-command shall cause the system to go to the System Command level from the PARAMETER mode. The system shall respond with the System Command prompt sequence.

3.8.7.4.2 EDIT Command. The system shall allow the operator to edit a selected operator entered message at any time. The system shall enter the Edit Mode and prompt with the Edit prompt sequence. If the operator does not enter a particular message number, the system shall assign one. The system shall assign these numbers sequentially, starting with 1 and ending with 9999. There shall never exist two messages with identical message numbers in the system. When 9999 is assigned to a message, the next number assigned will be the first free number starting back at 1. When the operator assigns a particular message number, the system shall check to see if this message exists and is a locally composed message in the system. If the requirements are met, an appropriate message stating the message number, followed by the EDIT prompt sequence shall be printed. If the requirements are not met, an error message shall be printed, followed by the System Command prompt sequence. Upon entering the EDIT mode, the line pointer shall point to the first line of the message and the EDIT Sub-commands shall be the only valid commands or sub-commands until the EXIT Sub-command is executed.

3.8.7.4.2.1 TOP Sub-command. The TOP Sub-command shall set the line pointer to the first line of the current message being edited and that line shall be printed. If the message is empty an error message shall be printed.

3.8.7.4.2.2 BOTTOM Sub-command. The BOTTOM Sub-command shall set the line pointer to the last line of the current message being edited and that line shall be printed. If the message is empty an error message shall be printed.

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3.8.7.4.2.3 UP Sub-command. The UP Sub-command shall move the line pointer the number of lines, as indicated by the operator, towards the beginning of the current message being edited and shall print the line at the new line pointer position. If the number of lines is not entered it shall be assumed to be 1. If an "*" is entered, the effect shall be the same as executing the TOP Sub-command. If the number of lines entered is greater than 99 or not numeric, an error message shall be printed. If the number is greater than the number of lines from the current line pointer position to the first line of the message, the line pointer shall be moved to the first line and that line shall be printed. If the current message being edited is null, an error message shall be printed.

3.8.7.4.2.4 DOWN Sub-command. The DOWN Sub-command shall move the line pointer the number of lines, as indicated by the operator, towards the end of the current message being edited and shall print the line at the new line pointer position. If the number of lines is not entered it shall be assumed to be 1. If an "*" is entered, the effect shall be the same as executing the BOTTOM Sub-command. If the number entered is greater than 99 or not numeric, an error message shall be printed. If the number is greater than the number of lines from the current line pointer position to the last line of the message, the line pointer shall be set at the last line of the message and that line shall be printed. If the current message being edited is null, an error message shall be printed.

3.8.7.4.2.5 BEFORE Sub-command. The BEFORE Sub-command shall allow the operator to insert a block of text before the current line. The process shall normally be terminated by depressing the HLT key. The line pointer shall reamin at the original line. The BEFORE Sub-command shall provide no prompt line. Lines shall be entered until the desired block has been entered. Depressing the HLT key shall then indicate that the desired text has been created and the sub-command shall be terminated. The actual insertion into the message shall be accomplished line by line. Each input line of text shall be

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entered into the message when a CR is entered or when the line reaches the current line length. In this latter case, a CR shall be automatically appended to the line. A non-empty line terminated by depressing the HLT key shall append a CR to the line, insert the line into the message, and terminate the BEFORE Sub-command. Depressing the HLT key with an empty line shall cause the sub-command to be terminated. If there is not sufficient memory available to insert the current line, an error message shall be printed and the current input line shall not be inserted into the message, and the sub-command shall be automatically terminated. Within the system, blank lines, i.e., lines which consist of only a CR, shall always be inserted into the memory as a SPACE code followed by a CR code.

3.8.7.4.2.6 INSERT Sub-command. The INSERT Sub-command shall allow the operator to insert a block of text after the current line. The command shall normally be terminated by depressing the HLT key. The line pointer shall be reset to point to the last inserted line of text. The INSERT Sub-command shall provide no prompt line. Lines shall be inserted until the desired block of text has been entered. Depressing the HLT key shall indicate that the desired block of text has been inserted and the command shall be terminated. Each line of text shall be inserted into the message either when a CR is entered or when the line reaches the current line length. In the latter case, a CR shall be appended to the line and the line inserted after the current line. The line pointer shall be reset to point to the last line inserted. A non-empty line terminated with HLT shall append a CR to the line, shall insert the line into the message, shall set the line pointer to the inserted line, and terminate the command. Depressing the HLT key with an empty line shall cause the command to be terminated. If there is insufficient memory available to insert the current input line, an error message shall be printed; the line causing the condition is not inserted into the message; and the insert process shall be terminated. Within the system, lines which consist of only CR's shall be inserted into memory as a SPACE code followed by a CR code.

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3.8.7.4.2.7 DELETE Sub-command. This sub-command shall allow the operator to delete a specified number of lines from the currently edited message starting at the current line pointer. The number of lines to be deleted shall be specified by the operator. One line shall be assumed if nothing is specified. If an "*" is specified or the number of lines specified is greater than the number of lines remaining in the message, the remainder of the message shall be deleted. At the completion of a delete, the line pointer shall be set at either the next non-deleted line of the message or the new last line of the message. An error message shall be printed if the result of the deletion is an empty message. If the message is not null after the deletion, the line at the current line pointer shall be printed. If the number of lines specified is greater than 99 or not numeric, an error message shall be printed.

3.8.7.4.2.8 FIND Sub-command. This sub-command shall allow the operator to input a phrase of up to 20 characters to be found in the message text. The system shall search the subsequent lines as required to find the first occurrence of the phrase from this point in the message. The line pointer shall be set to the line in which the phrase is found. The line shall be printed and the sub-command terminated. Input of the phrase shall be such that any printing delimiter may be used by the operator to delineate the characters of the phrase. The phrase may not contain CR or delimiter character and must contain at least one character or an error message shall be printed. If the phrase is not found, the line pointer shall not be changed, an error message shall be printed, and the sub-command terminated.

3.8.7.4.2.9 LIST Sub-command. This sub-command shall allow the operator to print a specified number of lines of the current message being edited, beginning at the current line pointer. The line pointer shall not be changed. If the number of lines to be printed is not entered, it shall be assumed to be 1. If an "*" is entered, the remainder of the message shall be printed beginning at the current line. If the number of lines is greater than 99 or is not numeric, an error message shall be printed. If the number

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of lines is greater than the number of lines from the current line to the end of the message, the remainder of the message shall be printed. Depressing the HLT key during printing shall cause the printing to stop and an error message shall be printed. If the message is empty, an error message shall be printed.

3.8.7.4.2.10 REPLACE Sub-command. The REPLACE sub-command shall allow the operator to find a phrase of 1 to 20 characters in the message text and to replace the phrase with a second phrase of 0 to 40 characters. The system shall search the current line and subsequent lines as required to find an occurrence of the first phrase. The line pointer shall be set to this line. The characters in the first phrase shall be replaced with the characters of the second phrase. If the replacement would cause a line length to be exceeded, a CR shall be inserted to maintain the designated length, thus creating a new line. The line pointer shall be set to the new line. The new line or lines shall be printed. If the first phrase cannot be found, an error message shall be printed and the sub-command terminated. Input of the phrase shall be such that any printing character may be used by the operator to delineate and separate the two phrases. Neither phrase may contain a CR or delimiter character and the first phrase must contain at least one character or an error message shall be printed. If this sub-command shall cause an entire line to be deleted, i.e., all characters, the system shall delete the CR and the line pointer shall be set as if the lines were deleted by the DELETE Sub-command. Printing shall also be as if the DELETE Sub-command had been excuted.

3.8.7.4.2.11 HEADER Sub-command. The HEADER Sub-command shall provide prompting or a JANAP 128 PLAINDRESS header format. If the current line length is greater than 69, an error message shall be printed and the command terminated. The prompting shall consist of printing a series of questions followed by a carriage return and line feed and accepting a series of inputs from the operator. The number and type of characters to be input are given in parenthesis at the end of each question. If the input does not conform in number or type, an error message shall be printed and

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the question repeated. Whenever data is being inputed, leading and trailing spaces shall be allowed. The correction keys (DLC, DLL, REV) and the CAPIT Sub-command shall be active. All numbers shall be entered with or without leading zeros as required. If the HALT key is activated during prompting, the prompting sequence shall be terminated and control shall be returned to the Edit Mode with no changes in the message. The Routing Indicator/Plain Language Address (RI/PLA) entry shall be defined as follows: (Note: b_o indictes that 0 or more blanks may be entered).

RI / PLA

b_o RI b_o / b_o PLA b_o CR

Where RI is 4 to 7 alpha characters and PLA is any text not containing a carriage return. For purposes of constructing the header, the answer to each question shall be considered as a variable which carries the same number as the question (i.e., question 1, channel indentifier is V1). The text of each of the questions shall be as follows: (Note: The questions 10 and 11 shall be asked and valid input responses shall be accepted until either their sum total is 50 or a null line is entered for each question).

1. ENTER CHANNEL IDENTIFIER (3A)

2. ENTER ORIGINATOR ROUTING INDICATOR/PLAIN LANGUAGE ADDRESS (RI/PLA)

3. ENTER MESSAGE PRECEDENCE (1A)

4. ENTER ORIGINATOR AND DESTINATION LANGUAGE AND MEDIA FORMAT (2A)

5. ENTER CLASSIFICATION (1A)

6. ENTER CONTENT INDICATOR/COMMUNICATION ACTION IDENTIFICATION (4AN)

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7. ENTER STATION SERIAL NUMBER (4N)

8. ENTER JULIAN DATE (3N)

9. ENTER TIME FILED (4N)

10. ENTER ACTION ROUTING INDICATOR/PLAIN LANGUAGE ADDRESS (RI/PLA)

11. ENTER INFORMATION ROUTING INDICATOR/PLAIN LANGUAGE ADDRESS (RI/PLA)

12 ENTER TIME OF DAY (ZULU TIME) (4N)

13. ENTER DAY MONTH AND YEAR (2N 3A 2N)

14. ENTER CHANNEL SEQUENCE NUMBER (3N)

The variables and their corresponding number and kind of value shall be as follows: (Note: Line one above corresponds to variable 1 (V1), line 2 to variable 2 (V2), etc.)

VI - 3 alpha

V2 - RI/PLA

V3 - 1 alpha must be: Z- flash

O-immediate

P- priority

R- routine

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V4 - 2 alpha

V5 - 1 alpha must be: U - unclas

E - unclas EFTO

C - confidential

S - secret

T - top secret

V6 - 4 alpha/numeric

V7 - 4 numeric

V8 - 3 numeric

V9 - 4 numeric

V10 - RI/PLA (may have multiple inputs)

V11 - RI/PLA (may have multiple inputs)

V12 - 4 numeric

V13 - (2 numeric) (3 alpha) (2 numeric)

V13A - 2 numeric V13B - 3 alpha V13C - 2 numeric

V14 - 3 numeric

The variables shall be formed into a JANAP 128 PLAINDRESS header and trailer as follows:

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If the system is operating in ASCII mode, the header shall be formed as follows and inserted before the body of the message. For Items 1 through 8 the symbol b_1 (representing one and only one space) is used to clarify the description of the printout.

1 V1 V14 CR

2 V3 V4 V5 V6 b_1 (RI from V2) b_1 V8 V9- (repeat V5 4 times)--(RI's from V10 and V11 separated by spaces or CR as required). CR

3 ZN (if V5 = U then R else Y) b_1 (repeat V5 five times) CR

4 V3 b_1 V13A V12 b_1 V13b_1 V13C CR

5 FM b_1 V2 (PLA ONLY) CR

6 TO b_1 (V10 each entry is separated by a CR) CR

7 INFO b_1 (V11 each is entry is separated by a CR) CR

8 BT CR

9 Based on V5, the classification is entered here written out as follows:

U - UNCLAS CR

E - UNCLAS E F T O CR

C - C O N F I D E N T I A L CR

S - S E C R E T CR

T - T O P S E C R E T CR

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The trailer of the message shall be as follows and appended after the body of the message:

BT CR

V7 CR LF LF LF LF LF LF LF

If the system is operating in the BAUDOT mode, the header shall be formed as follows, and inserted before the body of the message. The LTRS and FIGS codes indicated shall be inserted into the message for transmission only. For items 1 through 8 the symbol b_1 (representing one and only one space) is used to clarify the description of the printout.

1 V1∧V14_ CR

2 V3 V4 V5 V6 b_1 (RI from V2) ∧ V7b_1 V8 V9-_

(repeat V5 4 times)∧--_RI's from V10 and V11 separated by spaces or CR as required)∧._ CR

3 ZN (if V5 = U then R else Y) b_1 (repeat V5 five times) CR

4 V3b_1∧V13A V12_ Zb_1 V13Bb_1 ∧ V13C _ CR

5 FMb_1V2(PLA ONLY)CR{ CR if V2 ends in FIGS mode / - CR if V2 ends in LTRS mode }

6 TO b_1 (V10 each entry is separated by

{ _CR if V10 ends in FIGS mode / CR if V10 ends in LTRS mode }

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7 INFO b_1 (V11 each entry is separated by

{ __ CR if V11 ends in FIGS mode
CR if V11 ends in LTRS mode }

8 BT CR

9 Based on V5, the classification is entered here written out as follows:

U - UNCLAS CR

E - UNCLAS E F T O CR

C - C O N F I D E N T I A L CR

S - S E C R E T CR

T - T O P S E C R E T CR

The trailer of the message shall be as follows and appended after the body of the message:

BT CR

^#V7__ CR LF LF LF LF LF LF LF

The LTRS code in the trailer is stored within the message.

V6 plus the PLA portions of V2, V10 and V11 may be alphanumeric. The format above assumes these to be alphabetic. If any numeric characters are contained within these parameters, FIGS and LTRS codes shall be inserted within the parameters and preceding the immediately following parameter as in normal message text. A FIGS code shall always precede the "/" which separates the RI and PLA in V2, V10, and V11.

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3.8.7.4.2.12 APPEND Sub-command. This sub-command shall allow the operator to append the contents of a designated message to the end of the current message. If the message number of the message to be appended is not a message in the system, or is the presently edited message an error message shall be printed and the command terminated. The pointer shall be set to the last line of the appended material and that line shall be printed. If the message to be appended contains characters with parity error, these error characters shall be converted to spaces as the append is carried out. If the APPEND Command shall cause memory to become full, an error message shall be printed and the command shall not be carried out.

3.8.7.4.2.13 EXIT Sub-command. This sub-command shall cause the system to go to the System Command level from the EDIT mode. The system shall respond with the System Command prompt sequence.

3.8.7.4.3 PRINT Command. This command shall allow the operator to specify the number of a message to be printed. If the number specified is not a message in the system, an error message shall be printed and the command shall be complete. If the message number is given, the subject message shall be printed. The system shall excute 10 LF's, print the message, mark the message as printed and excute 10 LF's. The message shall not be deleted from message memory when printing is completed. If printing is stopped, an error message shall be printed and the command shall be complete. The stored message shall be marked as printed if it is completely printed. If an "*" is given, all received messages shall be printed in order, first-in first-out by precedence. The system shall look up the number of the oldest, highest precedence message, excute 10 LF's print the message, and delete the message. This action shall be repeated until no received messages remain, then the system shall excute 10 LF's and the command shall be complete. If the printing is stopped, a message shall be printed and the command shall be complete.

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3.8.7.4.4 TRANSMIT Command. This command shall allow the operator to specify the number of a message to be transmitted. If the number specified is not a composed message in the system, an error message shall be printed and the command shall be complete. The specified message shall be transmitted and printed at the same time. In the BAUDOT mode, the LTR's code shall be the first character transmitted in any message. The specified message shall be marked as being transmitted, and transmission shall be initiated. The system shall excute 10 LF's, print the message, mark the message as printed, excute 10 LF's, and the command shall be complete. If a previous transmission has not been completed, an error message shall be printed and the command terminated. If printing is stopped by the HLT key, a message shall be printed, but transmission shall continue until completion. If an empty message is specified an error message shall be printed and no transmission shall be done. If the "TRANSMIT XXXX WITH NNN" option is used, the NNN shall replace characters 4, 5 and 6 in the first line of the message. If characters 4, 5 and 6 are not numeric or do not exist, an ERROR MESSAGE shall be printed and the command terminated. Upon completion of transmission, the message shall be marked as having been transmitted.

3.8.7.4.5 REMOVE Command. This command shall allow the operator to specify the number of a message to be removed from message memory. If this specified number is not a message in the system, an error message shall be printed and the command shall be complete. If the message is a received message and has been marked as printed, it shall be deleted, a message shall be printed and the command shall be complete. If the message has not been printed, an error message shall be printed, no other action shall be taken and the command shall be complete. If the message is a composed message which has been transmitted, it shall be deleted, an error message shall be printed and the command shall be complete. If the message is marked as being in transmission, an error message shall be printed, no other action shall be taken and the command shall be complete. If the message has not been transmitted, an error message shall be printed. If the next command is to remove the same message number, it shall be deleted and an error message shall be printed.

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3.8.7.4.6 JUSTIFY Command. The specified message shall be reformatted. If the specified number is not a message in the system, an error message shall be printed and the command shall be complete. The specified message shall be line justified. The specified message shall be accessed, all LF's shall be removed and all CR's replaced with spaces. CR's shall be inserted in place of spaces to form maximum length lines which are less than or equal to the current line length. The whole message shall be processed. No words shall be split, the CR's shall be inserted in place of spaces only. If any sequence of characters which is greater than the line length does not have a space to change to CR, the CR shall be inserted so as to form a line equal to the line length.

3.8.7.4.7 QUERY Command. The system shall print out the following information:

SWITCH STATE = ICT, KSR or RO

OPERATION STATE = ICT, KSR or RO

OPERATING CAPACITY = FULL or DEGRADED

MODE = ASCII or BAUDOT

BAUD RATE = 45.5, 50, 75, 150, 300, 600, or 1200

STOP BITS = 1 or 2

END OF LINE OPTION = 0 to 8 ASCII CHARACTERS REPRESENTING END OF LINE SEQUENCE

SPACE OPTION = ON or OFF

LINE LENGTH = 40 to 80

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LINE FEED = 1 or 2

RECEIVE ENVELOPE OPTION = START RECEIVE ENVELOPE SEQUENCE AND STOP RECEIVE ENVELOPE SEQUENCE

TRANSMIT ENVELOPE OPTION = START TRANSMIT ENVELOPE SEQUENCE AND STOP TRANSMIT ENVELOPE SEQUENCE

In the BAUDOT mode, the following information shall be printed:

BELL OPTION = FIGURES S or FIGURES J

In the ASCII mode, the following information shall be printed:

PARITY OPTION = ODD, EVEN or INHIBIT

CAPITAL LETTER OPTION = ON or OFF

3.8.7.4.8 STATUS Command. This command shall allow the operator to request the status of specified message(s). If an invalid message number is requested an error message shall be printed and the command shall be complete. If no message number is specified, the status of the latest received message shall be given. The status of the specified message(s) shall be printed and shall include: the message number, type of message (received or composed), precedence (as described below), number of memory blocks used by the message, and whether a received message has been printed or a composed message has been transmitted. If all messages are specified, "*" shall be entered as operand. Then the status of the received messages shall be printed in order, first-in first-out by precedence. The status of the composed messages in order of message number shall then be printed and the command shall be complete.

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The precedence shall be defined as:

Character	Precedence	Character Printed
Z	FLASH	Z
O	IMMEDIATE	O
P	PRIORITY	P
R	ROUTINE	R
OTHER OR NOT DETERMINABLE	ROUTINE	R

For composed messages the precedence shall be redetermined when exit from Edit mode is performed. The command shall be completed by printing:

"MEMORY NN % FULL."

Where NN shall be a two digit number representing the percent of message memory in use.

3.8.7.4.9 TTY Command. This command shall cause a one line message to be transmitted. In the BAUDOT mode, the LTR's code is the first character transmitted in any message. When the command is given, the status of the transmitter shall be checked. If a composed message is being transmitted an error message shall be printed and the command shall be terminated. If a previous TTY message is being transmitted, operator inputs shall not be accepted. If the transmitter is free, a CR/LF shall be issued and the operator may enter the text of the line. The text of the line is composed of all characters entered (with DLC, DLL and REV Keys available) until a CR is entered. The line of text shall be transmitted with an end of line sequence appended and in an envelope. If there is no text, the command shall be terminated. The command shall then be complete.

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3.8.7.5 Full duplex operation. The equipment shall be capable of full duplex operation (transmit and receive simultaneously) in a four wire system.

3.8.8 Performance requirements.

3.8.8.1 Response times. The system response time for all operator inputs shall be such that the operator senses no perceptible delay between operator input and equipment outputs. The maximum processing and editing command response time shall be two seconds.

3.8.8.2 Storage capacity. The equipment hardware shall provide a message storage capacity of ten (10) pages of twenty (20) lines per page, eighty (80) characters per line. The message storage memory block shall be capable of being easily removed from the system without affecting its ability to print messages. Incoming messages shall not be lost or affected in any way due to the message storage having been removed. This condition shall be maintained as long as the data rate does not exceed the print speed capability of the system.

3.8.8.2.1 Memory retention. In the event of the loss of prime power, information in message memory shall be retained by providing back-up power from an external power source. The power source back-up shall be provided through a connector in the interface.

3.8.8.3 Printing speed. The equipment printing speed shall be not less than 60 characters per second when operating continuously.

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3.8.8.4 Print quality. Characters shall be printed at 12 characters per horizontal inch (80 characters ± 1/2 character per 6.67 inches) and 6 lines per vertical inch (66 lines ± 1 line per 11 vertical inches). Registration of printed characters shall be such that the centers of all characters are within ± 0.01 inches of the nominal center of the line.

3.8.8.5 Automatic time-out. The print drum motor shall be turned off if printing has not been required for 1± 1/4 minute. When printing is required there shall be no loss of printed characters.

3.8.9 Operator prompting.

3.8.9.1 Error/information message list. Information/error messages shall either be printed in full or by appropriate number.

3.8.10 Transfer switch. The transfer switch shall be wired between a spare pin of the communication connector and ground.

3.9 AN/UGC-74A(V) signal characteristics and electrical requirements.

3.9.1 Operating modes. This equipment shall operate in the following modes one at a time.

3.9.1.1 ASCII code interface.

a. Start/stop, balanced low level digital interface using internal clock.

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b. Start/stop, balanced low level digital interface using an external clock.

c. Start/stop, high level digital interface using internal clock.

d. Start/stop, high level digital interface using an external clock.

3.9.1.1.1 Signal speeds(ASCII). The equipment shall operate with signals at speeds (bps) as follows:

a. ASCII 10.0 unit, start/stop at 75, 150, 300, 600, and 1200. (Balanced low level).

b. ASCII 10.0 unit, start/stop at 75 and 150 (High level).

c. ASCII 11.0 unit, start/stop at 75, 150, 300, 600, and 1200. (Balanced low level),

d. ASCII 11.0 unit, start/stop at 75 and 150 (High level).

3.9.1.1.2 External clock mode. This equipment shall operate up to 1200 bps (Balanced low level digital interface); 75 and 150 bps (high level digital interface).

3.9.1.2 BAUDOT code interface.

a. Start/stop, balanced low level digital interface using internal clock.

b. Start/stop, balanced low level digital interface using external clock.

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3.9.3.1 Signals. The equipment shall provide nonreturn to zero (NRZ) and conditioned diphase signals in accordance with figure 2. The type of signal shall be selectable by means of an internal switch.

3.9.3.1.1 Conditioned diphase. The conditioned diphase signal shall be provided only for data rates of 300 bps , 600 bps and 1200 bps.

3.9.3.2 High level digital signal interface.

3.9.3.2.1 Idle transmit condition. The transmit interface shall provide a closed circuit for mark and an open circuit for space for a 130 V (±2%), 60 or 20 mA signal line.

3.9.3.2.2 Transmitter source impedance. The source impedance of the transmit interface shall not exceed 150 ohms.

3.9.3.2.3 Receive interface (130-volt operation), The receive interface shall be insensitive to the polarity of the receive signal in the 130V (±2%), 60 or 20 mA signal line. A change from 32 mA to 28 mA on a 60 mA neutral signal and from 12 mA to 8 mA on a 20 mA neutral signal shall cause correct operation of the receive interface.

3.9.3.3 Normal input keying (NIK) interface. The equipment shall be compatible with the LOOP IN and LOOP OUT circuits of COMSEC equipments such as the KW-7. The RECEIVE circuit shall simulate the characteristics of a teletypewriter selector magnet when the driving element produces an output current of 20 mA and open circuit voltage of 48V. The SEND circuit shall provide a switch closure for mark and an open for space. The mark loop current supplied by the COMSEC equipment shall be approximately 70 microamperes. The maximum value of shunt capacitance to ground at the COMSEC equipments LOOP INPUT is defined to be 0.07

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microfarads, and the mark to space decision is made at an input voltage of +1.5V. This voltage shall be defined as the point where the NIK wave-shape is sampled for all distortion measurements. The maximum allowable distortion under these conditions shall be 35 percent for marking and spacing end or characteritic distortion (see 6.4.2) and 40 percent for marking or spacing bias distortion. These distortion limits supersede the requirements of 3.9.4.2 for NIK operation only.

3.9.4 Signal distortion.

3.9.4.1 Receive equipment. When internal clock is selected, the receiving equipment shall operate error-free when the received data signal is degraded as described in 3.9.4.1.1 or 3.9.4.1.2. The equipment shall operate error-free when these degradations occur simultaneously, when neither one degradation is more than 50 percent of its indicated maximum.

3.9.4.1.1 Digital distortion. The maximum bias or end distortion of the received low-level data signal will be 42 percent of the shortest duration pulse encountered in the normal signal. That is, maximum digital distortion for non-return to zero formatted data shall be 42 percent of a bit period and for conditioned diphase formatted data shall be 21 percent of a bit period. The maximum bias or end distortion of the received current-loop data signal shall be 42 percent of the shortest duration pulse encountered in the nominal signal.

3.9.4.1.2 Spurious pulses. The received data signal may contain spurious pulses having a time duration less than 25 ± 1 percent of the shortest duration pulse encountered in the nominal signal. Rejection of individual spurious pulses separated by less than 10 percent of a nominal bit period shall not be required if their combined pulse width exceeds the 25% requirements. This requirement shall not apply when using external clock.

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3.9.4.2 Transmit equipment. The transmitting equipment output data signals shall have less than two percent bias or end distortion and shall be at the specified bit rate ± 0.1 percent.

3.9.5 Isolation of chassis ground, signal, and clock lines. The lines shall be isolated from chassis ground, and the third line in the cables shall be grounded to the external cable shield.

3.9.6 System interface. The equipment shall operate satisfactorily with the KW-7 (interface defined in 3.9.3.3).

3.9.6.1. Cables. Cables to interconnect the AN/UGC-74A(V) and the equipments listed in 3.9.6 shall be provided by the contractor for test purposes only. Drawings for the construction of the interconnecting cables shall be provided by the contractor.

3.9.7 Power.

3.9.7.1 Power consumption. The steady state power drain shall not exceed 100 watts.

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3.9.7.2 Sources. The equipment shall be capable of operation from the following power sources by only changing external power cords:

a. 22 to 30 VDC vehicular electrical system (nominal 26 ± 4 VDC).

b. 115/230 V, 50/60/400 cycle, single phase, AC. AC voltage tolerances of ± 15 percent and frequency tolerance of ± 5 percent.

3.9.7.3 Over voltage and transients.

3.9.7.3.1 AC overload. The equipment shall be capable of with-standing an AC input voltage 20 percent greater than the specified high line voltage for a five minute period, with no component degradation. The AC over voltage shall not cause a loss or degradation of date.

3.9.7.3.2 DC over voltage and transient voltage protection. The equipment shall operate without degradation of equipment or data during over voltage and transient voltages in vehicular power supplies as follows:

a. Over voltage 36 VDC input for a 1-hour maximum.

b. Over voltage transient ± 250 V peak spike induced on the normal input voltage having a rise time not exceeding 50 ns, a frequency of oscillation greater than 100 kHz and less than 600 kHz and an energy content of 15 millijoules maximum. Data degradation is permissible during each transient.

c. Under voltage transients, 15 VDC input. Data degradation permissible during each transient.

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3.9.7.4 Isolation of chassis ground, power lines. The grounded circuit conductor (power neutral) shall be isolated and insulated from the equipment grounding conductor (chassis ground). A separate line shall be used to carry the equipment grounding conductor. The only exception to the preceding is that the equipment grounding conductor shall be connected to the grounded circuit conductor at the power source.

3.10 Keyboard Transmitter TT-745/UGC-74A(V) requirements.

3.10.1 Operating modes. The keyboard is a dependent device requiring a Teleprinter TT-744/UGC-74A(V) operation. The keyboard shall conform to the applicable operating modes of 3.9.1 and 3.9.2.

3.10.2 Keyboard arrangement. The keyboard shall consist of the 59 standard keys required by MIL-STD-1280 for type 1, Class 1 keyboards plus four additional keys for basic editing control together with the supporting structure for these 63 keys.

3.10.3 Two key rollover. The two key rollover feature shall prevent loss or incorrect generation of keyboard code output when a second key is depressed before the release of a first key. The code of the second key shall be generated upon the release of the first key.

3.10.3.1 Key pressure. The pressure necessary to operate any key shall not be less than 3 ounces or more than 6 ounces. The required striking force shall be the same at any speed and for any equipment tilted up to 20 degrees form normal (10% sampling).

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3.10.4 Keyboard.

3.10.4.1 Lower case characters. The lower characters shown on the key-tops shall be the characters processed with keyboard in normal state, i.e., neither the shift nor control key operated.

3.10.4.2 Shift key. The upper characters shown on the keytops shall be the characters processed with the shift key operated.

3.10.4.3 Shift lock key. A depression of the LOC key (shift lock key) shall lock the SHIFT key in its operating mode. Momentary depression of the SHIFT key shall disengage the LOC key and return the keyboard to its normal state. An indication shall be given whenever the keyboard is in the shift position,

3.10.4.4 Machine functions. The characters indicated on the following keytops shall be processed, regardless of control and/or shift key positions. CR (carriage return), LF (line feed), BS (back space), DEL (delete), TAB (horizontal tab), and SPACE.

3.10.4.5 Control codes. The two and three character groups (control characters) shown on the left side of the key tops shall be processed when the control key is operated regardless of the position of the shift key.

3.10.4.6 Control keys. The control key shall have no effect on the codes generated from keys with no control character groups on the key tops.

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3.10.4.7 Repeat key. When the RPT key is operated, the code corresponding to that of the last coded key operated shall be generated continuously until the RPT key is released. The character rate for the repeat operation shall be five characters per second.

3.10.4.8 TAB key. When the TAB key is pressed, the print position shall be moved to the smallest tab position value which is greater than the print position. The next character shall be printed in that position. If the present print position is greater than or equal to the largest tab position, then the tab shall be replaced by a CR. If the space option is set, then the tab function shall be carried out by inserting the proper number of spaces into the buffer instead of the tab character. If the space option is not set, then the tab character is put in the buffer and the print position shall be set.

3.10.4.9 Message processing keyboard. The LTRS and FIGS codes shall be individually entered by depressing the DEL or BS key respectively (Applicable only in BAUDOT mode). The following four special keys on the keyboard shall be used for basic editing on all lines of input into the terminal, command or otherwise:

Key	Description
DLC	Shall delete the last character input. The system response shall be a backward slash over the character to be deleted to indicate the character has been deleted. This key shall only be valid over the current line. If the current line is empty a CR/LF shall be outputted.

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DLL	Shall delete the current input line. The system response shall be a carriage return, line feed, to indicate that the current input line has been deleted. This key shall only be valid over the current input line before a CR is entered. If the current line is empty a CR/LF shall be outputted.
REV	Shall review or reprint the current input line. The system shall respond with a carriage return, line feed, and a reprint of the current input line. The next character entered shall be printed immediately following the last character of the reviewed line. This key shall only be valid over the current line. If the current line is empty a CR/LF shall be outputted.
HLT	Shall halt the operator initiated printing currently being performed. The system shall respond by terminating the printing and follow with CR/LF, "PRINT ABORTED" CR/LF and the proper prompt sequence. The HLT key shall also be used to terminate the INSERT and BEFORE Sub-command.

3.11 Tempest requirements. The tempest characteristics of the equipments shall be in accordance with the requirements of MIL-T-49261.

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3.12 Acoustic noise. Acoustic noise shall be sufficiently low in intensity to comply with the noise level requirements of MIL-STD-1474 as modified in Table IV.

3.13 Electromagnetic compatibility. The equipment shall comply with the following requirements of MIL-STD-461A Notice 4:

a. First article group D inspection.

CE01	CS01	RE02 B/	RS02
CE02	CS02	RE02.1 A/	RS03 C/
CE03	CS06	RS03.1A/ C/	
CE04 B/			
CE05			

A/ The upper frequency limit for radiated requirements RE02.1 and RS03.1 shall be 1,000 MHz.

B/ Transient interference due to initial on/off power transients and line break are exempt from meeting the requirements.

C/ The E-Field test signal shall be amplitude modulated 50% with 1 KHz tone as follows:

Frequency (Hz)	E-Field
10k -2M	1 Volt per Meter
2M -30M	5 Volt per Meter
30M -1.000M	10 Volt per Meter

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TABLE IV. Acoustic noise levels.

OCTAVE BAND * CENTER FREQ (Hz)	OCTAVE BAND CENTER FREQ (Hz)	NOISE LEVEL IN DB**
53	63	79
106	125	73
212	250	71
425	500	69
850	1000	67
1700	2000	65
3400	4000	62
6400	8000	60

* THE REASON FOR THE TWO OCTAVE BANDS IS TO PERMIT THE USE OF AVAILABLE TEST EQUIPMENT WHICH WOULD NORMALLY FALL INTO ONE OF THE TWO OCTAVE BANDS.

** REFERENCED TO 0z0002 DYNES/CM^2

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b. Subsequent Group D inspection.

CE04	B/	RE02	A/	B/
CE05		RE02,1	A/	

A/ The upper frequency limit for RE02 and RE02.1 shall be 400 MHz.

B/ Transient interference due to initial on/off power transients and line break are exempt from meeting the requirements.

3.14 System nuclear survivability. All contractor furnished mechanical configurations, electronic assemblages, electronic equipment, electronic circuit and electronic pieceparts shall withstand the nuclear environment as specified in EL-CP0000-0031 and appendix 1, dated 14 November 1969 (Blast and thermal not applicable). The equipment shall meet the operational requirements of section 3 herein, after being subjected to the specified nuclear environments. This recovery in operational effectiveness may include automatic or manual correction of nuclear-environment-induced changes in the system but not the replacement of pieceparts or components. The equipment shall be energized and the CFE and GFE configured for the worst case system response during exposure. The contractor shall ensure that the GFE is not degraded by nuclear environment-induced signals coupled from the CFE; and vice versa, that the CFE is not degraded by nuclear environment-induced signals coupled from the GFE.

3.15 Environmental service conditions. The equipment shall be constructed so that no fixed part shall become loose, cracked, flaked, pitted or blistered, no movable part or control shall be shifted in setting, position or adjustment, and that there shall be no permanent degradation in performance under the following environmental conditions.

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3.15.1 Temperature.

a. Operating. Ambient temperature in the range -25 degrees F(-31.7 degrees C) to + 155 degrees F (+68 degrees C) (+120 degrees F (+49 degrees C) plus the heating effects of solar radiation at the rate of 360 BTU/ft^2/hr.). In the low temperature range (between -25 degrees F (-31.7 degrees C) and +20 degrees F (-6.7 degrees C)) the following is permissable:

(1) Special ribbons

(2) Operator assistance for paper motion during the first two minutes of operation.

b. Non-operating. Exposure in the range -65 degrees F (-54 degrees C) to +155 degrees F (+68 degrees C).

3.15.2 Humidity. Humidities of 94% +4 % over a temperature range of +68 degrees F (+20 degrees C) to +149 degrees F (+65 degrees C)/

3.15.3 Altitude.

a. Operating. Elevations up to 10,000 feet above sea level.

b. Non-operating. Elevations up to 17,500 above sea level.

3.15.4 Immersion. The equipment shall be immersed to a covering depth of three feet of water for a period of two hours without evidence of any water leakage.

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3.15.5 Vibration.

3.15.5.1 Resonance search. The equipment shall have no resonances in the frequency range of 10 up to 55 Hz that exceed twice the amplitude of the applied vibration. Parts or assemblies which are designed to move, rotate, pivot or otherwise move in order to fulfill other contractual requirements can exceed twice the input vibration amplitude. However, such parts shall not sustain damage as a result of the resonant search and equipment performance shall not be degraded.

3.15.5.2 Bounce, vehicular. The equipment shall correctly receive and transmit data in any army vehicle while that vehicle is in motion across fields and hard roads. The equipment, as mounted for field use, shall withstand 10g shock input (operating).

3.15.5.3 Bounce, loose cargo. The equipment, including necessary paper supply (full roll), shall withstand the shocks normally induced during field transportation as loose cargo.

3.15.6 Shock.

3.15.6.1 Drop. The equipment shall withstand shocks normally encountered during loading and unloading of equipments. After normal setup procedures, the equipment must meet all requirements.

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3.15.6.2 Bench handling. The equipment shall withstand shock encountered in servicing.

3.15.7 Ballistic shock. The equipment as mounted for operation with power supplied shall withstand simulated ballistic shocks without any part becoming separated from the whole. Equipment operational malfunctions during and up to one second after impact are permissible. Both hardware and software integrity shall be maintained. The equipment shall be completely operational following each shock and shall not require any operator intervention (e.g. manual re-initialization) to maintain its pre-test operational posture.

3.15.8 Dust. The equipment shall operate after being in the operating mode in blowing dust, as encountered in the desert.

3.15.9 Rain. The equipment shall be operable without degradation in specified performance and sustain no physical damage during periods of heavy precipitation with intermittent wind. It is allowable to replace paper and/or ink ribbons after the test. Operator assistance as required is allowed for paper movement during operation of equipment.

3.15.10 Salt fog. The equipment shall be resistant to the corrosive effects of salt fog atmosphere.

3.15.11 Orientation. The equipment, bolted in place to the mounting surface, shall operate in any orientation up to 20 degrees from normal operating position (that is forward, backward, left or right).

3.15.12 Fungus. The equipment, in the assembled and ready for delivery condition, shall provide no nutrients in material, coating, or contaminant form or support fungal growth when tested in accordance with paragraph 4.15.12. Only inherently fungus resistant grades of materials per requirement 4, MIL-STD-454, shall be used.

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3.15.13 Leakage air seal test. The equipment shall lose no more than 0.01 pound per square inch in one minute when a vacuum of 1 pound per square inch is provided in the equipment.

3.16 Reliability. The quantitative requirements of the AN/UGC-74A(V) with micro-processor and keyboard shall have a specified (0_o) mean-time-between-failure (MTBF), as defined in MIL-STD-781, of 1800 hours.

3.17 Systems safety engineering.

3.17.1 Personnel hazards, Personnel hazards shall be kept to a minimum. It shall be possible to change or replenish the printer paper or ribbons without exposing the operator to surfaces 35 degrees F above ambient, electrical potentials in excess of 30V, and motor driven mechanisms. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual inspection (see 4.22).

3.17.2 Edge rounding exposed. Exposed edges shall be rounded sufficiently to minimize laceration puncture hazards. The goal being a minimum radius of 0.04 inch (1 mm), and exposed corners to a minimum of 0.5 inch (13mm).

3.17.3 Radioactive materials. Radioactive materials shall not be used (e.g. luminous dials/markings, electron tubes, surge arrestors and lenses).

3.17.4 Overload protection. Electrical overload protection shall be in accordance with Requirement 1 and 8 of MIL-STD-454,

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3.18 Maintainability.

3.18.1 Scheduled maintenance. The equipment shall be capable of operation for at least 500 hours between lubricants periods or before adjustments are required to compensate for degradation of performance. It shall be capable of operation for at least 1000 hours, observing presecribed maintenance and lubrication schedules before replacement of any part is required.

3.18.2 Consumables. The time required to replace consumables, assuming no repaires are required, shall not exceed five minutes.

3.18.3 Mean-time-to-repair. Mean down time allowable to return the equipment to service shall not exceed four hours for scheduled maintenance servicing. Mean-time-to-repair (MTTR) is defined as the total corrective maintenace time divided by the total number of corrective maintenance actions during a given period of time. MTTR applies to the replacement of all modules, (see 6.4.11) subassemblies and chassis mounted piece parts. It does not include the repair of modules and subassemblies. The MTTR (M_C) shall be 30 minutes at the orginizational level and 2-1/2 hours at the direct support level.

3.19 Weight. The weight of the equipment shall not exceed 103 pounds.

3.20 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

a. In MIL-P-11268

Plastic material and parts.

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Wiring and cabling, including:

Protection
Clearance
Splicing and stretching
Grounding
Wire and cable shielding
Cleaning
Soldering and brazing
Welding
Controls

b. In this specification:

3.2 Construction

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).

b. Inspection covered by subdsidiary documents (see 4.4).

c. Quality conformance inspection

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Quality conformance inspection of packaging (see 4.18).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

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4.3.1 First article units. The quantities shall be as cited in the bid request and contract.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, preconditioning specified in 4.6., and the inspections specified for software verification, group A, group B, group C and group D (see tables V, VII, VIII, IX and X respectively).

4.3.2.1 First article software verification inspection. Two units shall be subjected to the software verification inspections in accordance with table V. These units shall be selected from the first lot of first article units. These units shall have passed group A and group B inspecitons.

4.3.2.2 First article group A inspection. All first article units shall be subjected to the group A inspection (see table VII).

4.3.2.3 First article group B inspecton. The first article group B inspection shall be performed on units from the first article lot (see table VIII).

4.3.2.4 First article group C inspection. First article units shall be subjected to group C inspection in accordance with table VI. The selected units shall have passed group A and group B inspections.

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4.3.2.5 First article group D inspection. First article units shall be subjected to group D inspection (see table X). The selected units shall have passed group A and group B inspections. The reliability of the equipment shall be demonstrated in accordance with Appendix B, Category 1, 50.1 of MIL-STD-781C. The number of randomly selected equipment to be tested shall be eight (8). The specified MTBT (θ_O) to be used is located in 3.16.

4.3.3 First article data. The first article test plan, inspection acceptance procedure, and test report shall be as required in the contract or purchase order.

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TABLE V. Software verification inspection 1/

REQUIREMENT	REQ'T PARA.	INSP. PARA.
Coded BAUDOT Printer Functions	3.8.5.2	4.9.5.2
Carriage Return	3.8.5.2.1	4.9.5.2.1
Line Feed	3.8.5.2.2	4.9.5.2.2
Letters	3.8.5.2.3	4.9.5.2.3
Figures	3.8.5.2.4	4.9.5.2.4
Keyboard/Print Codes	3.8.5.3.1	4.9.5.3.1
TAB	3.8.5.3.2	4.9.5.3.2
CR	3.8.5.3.3	4.9.5.3.3
LF	3.8.5.3.4	4.9.5.3.4
Automatic Carriage Return	3.8.5.4	4.9.5.4
ICT State	3.8.6.4	4.9.6.4
RO State	3.8.6.5	4.9.6.5
KSR State	3.8.6.6	4.9.6.6.a & d
Self Test State	3.8.6.8	4.9.6.8.c
Message Processing	3.8.7	4.9.7
Message Storage	3.8.7.1	4.9.7.1
Memory Full	3.8.7.1.1	4.9.7.1.1
Operator Input	3.8.7.1.1.1	4.9.7.1.1.1
Incoming Message	3.8.7.1.1.2	4.9.7.1.1.2
Message Receipt	3.8.7.2	4.9.7.2
Receive Message Process	3.8.7.2.1	4.9.7.2.1
Transmit Message Process	3.8.7.3	4.9.7.3
System Commands	3.8.7.4	4.9.7.4
PARAMETER Command	3.8.7.4.1	4.9.7.4.1
LINE Sub-command	3.8.7.4.1.1	4.9.7.4.1.1
SETAB Sub-command	3.8.7.4.1.2	4.9.7.4.1.2
CLRTAB Sub-command	3.8.7.4.1.3	4.9.7.4.1.3
DISPLAY Sub-command	3.8.7.4.1.4	4.9.7.4.1.4
CAPIT Sub-command	3.8.7.4.1.5	4.9.7.4.1.5
REMIND Sub-command	3.8.7.4.1.6	4.9.7.4.1.6
EOL Sub-command	3.8.7.4.1.7	4.9.7.4.1.7
LF Sub-command	3.8.7.4.1.8	4.9.7.4.1.8
SPACE Sub-command	3.8.7.4.1.9	4.9.7.4.1.9
PAPER Sub-command	3.8.7.4.1.10	4.9.7.4.1.10
RENVELOPE Sub-command	3.8.7.4.1.11	4.9.7.4.1.11

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TABLE V. Software verification inspection 1/ (contd)

REQUIREMENT	REQ'T. PARA.	INSP. PARA.
TENVELOPE	3.8.7.4.1.12	4.9.7.4.1.12
EXIT Sub-command	3.8.7.4.1.13	4.9.7.4.1.13
EDIT Command	3.8.7.4.2	4.9.7.4.2
TOP Sub-command	3.8.7.4.2.1	4.9.7.4.2.1
BOTTOM Sub-command	3.8.7.4.2.2	4.9.7.4.2.2
UP Sub-command	3.8.7.4.2.3	4.9.7.4.2.3
DOWN Sub-command	3.8.7.4.2.4	4.9.7.4.2.4
BEFORE Sub-command	3.8.7.4.2.5	4.9.7.4.2.5
INSERT Sub-command	3.8.7.4.2.6	4.9.7.4.2.6
DELETE Sub-command	3.8.7.4.2.7	4.9.7.4.2.7
FIND Sub-command	3.8.7.4.2.8	4.9.7.4.2.8
LIST Sub-command	3.8.7.4.2.9	4.9.7.4.2.9
REPLACE Sub-command	3.8.7.4.2.10	4.9.7.4.2.10
HEADER Sub-command	3.8.7.4.2.11	4.9.7.4.2.11
APPEND Sub-command	3.8.7.4.2.12	4.9.7.4.2.12
EXIT Sub-command	3.8.7.4.2.13	4.9.7.4.2.13
PRINT Command	3.8.7.4.3	4.9.7.4.3
TRANSMIT Command	3.8.7.4.4	4.9.7.4.4
REMOVE Command	3.8.7.4.5	4.9.7.4.5
JUSTIFY Command	3.8.7.4.6	4.9.7.4.6
QUERY Command	3.8.7.4.7	4.9.7.4.7
STATUS Command	3.8.7.4.8	4.9.7.4.8
TTY Command	3.8.7.4.9	4.9.7.4.9
Storage Capacity	3.8.8.2	4.9.8.2
Error/Informantion Message List	3.8.9.1	4.9.9.1

1/ This document does not outline the sequence of tests necessary to demostrate compliance with the requirements of the table.

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TABLE VI. First article environmental inspection

	SAMPLE UNIT 1/								
INSPECTION	1,2,3	4	5	6	7	8	9,10	11,12	13
Immerson	5,8								
AC overload	3								
Power Consumption	1								
Sources	2								
DC overload and transient	4								
Vibration (resonance search)	6								
Shock, drop	7								
Temperature high		1					1		
Temperature low		2					2		
Altitude		3							
Bench handling		4							
Bounce (loose cargo)		5					3		
Ballistic shock		6					4		
Humidity 2/			1					1	
Bounce (vehicular)			2					2	
Rain			3					3	
Orientation			4						
Acoustic noise			5					4	
Paper Usage			6					5	
Fungus 2/				1					
Salt Atmosphere 2/					1				
Maintainability Demonstration									1
Systems Interface						1			
Dust 2/						2			

1/ The numbers in the unit column specify the order of inspections for the indicated unit.

2/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspeciton.

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4.4 Inspection covered by subsidiary documents. The following shall be inspected under applicable subsidiary documents as part of the inspection before packaging:

Item	Where required
Marking	3.5
Parts, materials and Processes; General	3.3
Finish	3.4

4.5 Quality conformation inspection of equipment before packaging. Each unit of each lot of equipment shall be subjected to bounce preconditioning (see 4.6.1) and burn-in (4.6.2) in that order prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of bounce preconditioning shall be classified as a defect and recorded as such. The Government may, at its option, review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements (see 6.3). Test equipment for Government verification inspection shall be made available by the contractor.

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4.5.1 Group A inspection, Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table VII. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. The contractor shall subject each lot to sampling inspection, utilizing the procedures of MIL-STD-105, using the AQL's indicated in Table Vll.

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in an order satisfactory to the Government.

4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to Table VIII and to the special inspection levels of Table I of MIL-STD-105. The inspection level shall be S-4.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A and Group B inspection. The inspection shall consist of the inspections specified in Table IX. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for group C inspection.

4.5.3.1.1 Subgroup I. For this subgroup, one unit shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every month. Subgroup I tests shall not be run when Subgroup II is specified.

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TABLE VII. Group A inspection - AN/UGC-74A(V)

INSPECTION	REQUIREMENT	INSP. PARA.	AQL MAJOR	AQL MINOR
Visual & Mechanical	Keyboard Arrangement 3.10.2	4.11.2	1%	4%
	Workmanship 3.20	4.7	1%	4%
	Air Seal 3.15.13	4.15.13	1%	NONE

		REQ'T PARA.	INSP. PARA	AQL
Electro-mechanical	Visibility of Copy	3.8.2	4.9.2	
	Copy Illumination	3.8.3	4.9.3	
	Power Indicator	3.8.5.5.2	4.9.5.5.2	
	Self Test	3.8.5.6.5	4.9.5.6.5	
	External Power Switch	3.8.5.6.8	4.9.5.6.8	
	Cold Start State	3.8.6.2	4.9.6.2	2.5% for the group
	Operation Validation/ State Determination State	3.8.6.3	4.9.6.3.a.(1)	
	KSR State	3.8.6.6	4.9.6.6.b & c	
	Self Test State	3.8.6.8	4.9.6.8.a	

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TABLE VIII. Group B inspection - AN/UGC-74A(V)

REQUIREMENT	REQ'T PARA.	INSP. PARA.	AQL
Interchangeability	3.6	4.8	4%
ASCII/BAUDOT Control	3.8.4	4.9.4	4%
Local Printer Function	3.8.5.1	4.9.5.1	4%
Line Feed	3.8.5.1.1	4.9.5.1.1	4%
FIG S/J Switch	3.8.5.2.5	4.9.5.2.5	4%
Parity Error	3.8.5.3.5	4.9.5.3.5	4%
End of Line Signal	3.8.5.4.1	4.9.5.4.1	4%
Audible Alarm Adjustment	3.8.5.5.1.1	4.9.5.5.1.1	4%
Audible Alarm Actuation	3.8.5.5.1.2	4.9.5.5.1.2	4%
Battery Backup Indicator	3.8.5.5.2.1	4.9.5.5.2.1	4%
Paper Low Indicator	3.8.5.5.3	4.9.5.5.3	4%
Line Indicator	3.8.5.5.4	4.9.5.5.4	4%
Received Message Indicator	3.8.5.5.5	4.9.5.5.5	4%
Transmit Indicator	3.8.5.5.6	4.9.5.5.6	4%
Memory Full Warning Indicator	3.8.5.5.7	4.9.5.5.7	4%
Switches	3.8.5.6	4.9.5.6	4%
PARITY Switch	3.8.5.6.1	4.9.5.6.1	4%
STOP BITS Switch	3.8.5.6.2	4.9.5.6.2	4%
STATE Switch	3.8.5.6.3	4.9.5.6.3	4%
Data Inversion	3.8.5.6.4	4.9.5.6.4	4%
Abort Switch	3.8.5.6.6	4.9.5.6.6	4%
Audio Alarm Reset	3.8.5.6.7	4.9.5.6.7	4%
Operation Validation/State Determination State	3.8.6.3	4.9.6.3b	4%
Battery Backup State	3.8.6.7	4.9.6.7	2.5%
Self Test State	3.8.6.8	4.9.6.8b	2.5%
Full Duplex Operation	3.8.7.5	4.9.7.5	4%
Response Times	3.8.8.1	4.9.8.1	4%
Memory Retention	3.8.8.2.1	4.9.8.2.1	4%
Printing Speed	3.8.8.3	4.9.8.3	4%
Print Quality	3.8.8.4	4.9.8.4	4%
Automatic Time-Out	3.8.8.5	4.9.8.5	4%
Transfer Switch	3.8.10	4.9.10	4%
ASCII Code Interface	3.9.1.1	4.10.1.1	4%
Signal Speeds (ASCII)	3.9.1.1.1	4.10.1.1.1	4%
External Clock Mode	3.9.1.1.2	4.10.1.1.2	4%

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TABLE VIII. Group B inspection -AN/UGC-74A(V) - Continued

REQUIREMENT	REQ'T PARA.	INSP. PARA.	AQL
BAUDOT Code Interface	3.9.1.2	4.10.1.2	4%
Signal Speeds (BAUDOT)	3.9.1.2.1	4.10.1.2.1	4%
Speed Accuracy	3.9.1.3	4.10.1.3	4%
Clock INT/EXT/KG-30 Switch	3.9.2	4.10.2	4%
Internal Clock Mode - Transmit Clock	3.9.2.1	4.10.2.1	4%
External Clock/ Transmitted Data Relationship	3.9.2.2.1	4.10.2.2.1	4%
Data Transmission	3.9.2.3	4.10.2.3	4%
Detailed Requirements for Digital Interface Circuits 1/	3.9.3	4.10.3	4%
Signals	3.9.3.1	4.10.3.1	4%
Conditioned Diphase	3.9.3.1.1	4.10.3.1.1	4%
Idle Transmit Condition	3.9.3.2.1	4.10.3.2.1	4%
Transmitter Source Impedance	3.9.3.2.2	4.10.3.2.2	4%
Receive Interface (130 Volt Operation)	3.9.3.2.3	4.10.3.2.3	4%
Normal Input Keying (NIK) Interface	3.9.3.3	4.10.3.3	4%
Receive Equipment	3.9.4.1	4.10.4.1	4%
Digital Distortion	3.9.4.1.1	4.10.4.1.1	4%
Spurious Pulses	3.9.4.1.2	4.10.4.1.2	4%
Transmit Equipment	3.9.4.2	4.10.4.2	4%
Isolation of Chassis Ground, Signal, and Clock Lines	3.9.5	4.10.5	2.5%
Isolation of Chassis Ground, Power Lines	3.9.7.4	4.10.7.4	2.5%
Two Key Rollover	3.10.3	4.11.3	4%
Key Pressure	3.10.3.1	4.11.3.1	4%
Lower Case Characters	3.10.4.1	4.11.4.1	4%
Shift Key	3.10.4.2	4.11.4.2	4%
Shift Lock Key	3.10.4.3	4.11.4.3	4%
Machine Functions	3.10.4.4	4.11.4.4	4%
Control Codes	3.10.4.5	4.11.4.5	4%
Control Keys	3.10.4.6	4.11.4.6	4%
Repeat Key	3.10.4.7	4.11.4.7	4%
TAB Key	3.10.4.8	4.11.4.8	4%

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TABLE VIII. Group B inspection - AN/UGC-74A(V) - Continued

REQUIREMENT	REQ'T PARA.	INSP. PARA.	AQL
Message Processing Keyboard	3.10.4.9	4.11.4.9	4%
Safety 1/	3.17 through 3.17.4	4.22	2.5%

1/ First Article Test Only

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TABLE IX. Group C inspection - AN/UGC-74A(V)

REQUIREMENT	REQ'T PARA.	INSP. DATA
Subgroup I		
Immersion	3.15.4	4.15.4
Subgroup II		
Unit 1		
Temperature High	3.15.1	4.15.1.1
Temperature Low	3.15.1	4.15.1.2
Altitude	3.15.3	4.15.3
Bounce, Loose Cargo	3.15.5.3	4.15.5.3
Bench Handling	3.15.6.2	4.15.6.2
Ballistic Shock	3.15.7	4.15.7
Unit 2		
Power Consumption	3.9.7.1	4.10.7.1
Sources	3.9.7.2	4.10.7.2
Overvoltage and Transients	3.9.7.3	4.10.7.3
AC Overload	3.9.7.3.1	4.10.7.3.1
Immersion	3.15.4	4.15.4
Vibration (resonance search)	3.15.5.1	4.15.5.1
Shock Drop	3.15.6.1	4.15.6.1
Immersion	3.15.4	4.15.4
Subgroup III		
Humidity	3.15.2	4.15.2
Bounce (vehicular)	3.15.5.2	4.15.5.2
Rain	3.15.9	4.15.9
Orientation	3.15.11	4.15.11
Acoustic Noise	3.12	4.17
Paper Usage	3.8.1	4.9.1
Subgroup IV		
Unit 1		
Fungus	3.15.12	4.15.12

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TABLE IX. Group C inspection - AN/UGC-74A(V) - Continued

REQUIREMENT	REQ'T PARA.	INSP. DATA.
Unit 2 Salt Atmosphere	3.15.10	4.15.10
Unit 3 Dust	3.15.8	4.15.8
Subgroup V		
System Interface	3.9.6	4.10.6
Subgroup VI		
Software Verification	(See Table V)	

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4.5.3.1.2 Subgroup II. For this subgroup, two units shall be selected from the first production lot,. For subsequent group C inspection, two units shall be selected and tested in the sixth month, in the tenth month, and every six months thereafter. Immersion test shall be performed both before and after conduction of the vibration and shock drop tests.

4.5.3.1.3 Subgroup III. For this subgroup, one unit shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected and tested at six month intervals.

4.5.3.1.4 Subgroup IV. For this subgroup, three units shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected and tested for fungus one time at midpoint of the production cycle.

4.5.3.1.5 Subgroup V. For this subgroup, two units shall be selected from the initial production lot.

4.5.3.1.6 Subgroup VI. For this subgroup, two units shall be selected from the first production lot. For subsequent group C inspection, two units shall be selected every three months or 700 units, whichever occurs first.

4.5.3.2 Order of inspection within group C. Group C inspection shall be performed in an order which is satisfactory to the Government.

4.5.3.3 Group C failures. Actions required relative to group C failures shall be as specified in the contract or purchase order.

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4.5.4 Group D inspection. This inspection shall consist of the tests specified in table X and shall be performed on sample units from lots that have been subjected to and met group A and group B inspection. The reliability acceptance test shall be conducted every other month, starting in the first month of production and using Test Plan VIIC of MIL-STD-781C. The minimum number of randomly selected equipment to be tested shall be eight and th maximum number shall be twelve from each two months production. The sample (8 minimum, 12 maximum) shall be divided approximately even between units randomly selected from each of the two consecutive months production. The specified MTBF (0_o) shall be as specified in 3.16.

TABLE X. Group D inspection - AN/UGC-74A(V)

REQUIREMENTS	REQ'T PARA	INSP PARA
Undersired signal data transmissions (TEMPEST)	3.11	4.12
Electromagnetic compatibility	3.13	4.13
Nuclear	3.14	4.14
Reliability	3.16	4.16

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4.5.5 Reinspection of conforming group C and group D sample units. Unless otherwise specified, sample units which have been subjected to and pass group C or D inspection, as appropriate, may be accepted on contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Preconditioning.

4.6.1 Bounce preconditioning. Each equipment produced shall with shock mounts (if any) removed or blocked, be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester,shafts in phase shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of three inches as the unit bounces. The unit shall be subjected to this preconditioning for one minute.

4.6.2 Burn-in debugging.

4.6.2.1 Test conditions. Every equipment shall be subjected to a 42 hour burn-in period. The last 24 hours of burn-in shall be failure free. Failure criteria is to be determined by 4.16.9. Each unit which fails during the 24 hours failure free period shall be repaired and returned to test until it successfully survives a 24 hour period without failure. In addition, all spare modules produced shall be subjected to the same test conditions (above). The test environment utilized during burn-in shall be IAW Appendix B, Category 1, 50.1 of MIL-STD-781C. Proper ribbon lift operation (if so equipped) shall be verified once each hour.

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4.6.2.2 Compliance. Units passing burn-in debugging shall be submitted to group A, B and either C or D inspection in that order without benefit of repair, cleaning, realignment, or other changes, unless specified herein. If after successful completion of burn-in debugging the equipment fails either the group A or B tests, the equipment will be reworked and resubmitted to burn-in debugging.

4.7 Visual and mechanical inspection of equipment. Equipment shall be examined for the defects listed in MIL-STD-252. (see 3.20).

4.8 Inspection for dimensional interchangeability. The dimensions listed below shall be gauged to determine conformance to the physical interchangeability requirement of 3.6. When a listed dimension is not within specified limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

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b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separate parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts on the plug-in assembly or major unit.

4.9 Functional requirements for Terminal, Communications AN/UGC-74A(V). Within the following subparagraphs, unless stated otherwise, all tests are to be performed from an initial state, not necessarily sequentially.

4.9.1 Paper usage tests. Install a 5-inch roll of single copy paper in the printer arranged for friction feeding. With power on, depress and hold the printer LINE FEED switch for at least one second while observing that the paper feeds to determine conformance to 3.8.1. Repeat for a 5-inch roll of multi-ply paper. Also, print out a minimum of three lines for each test. Paper used shall be friction per UU-P-547, Type 1, Class 1, Grade B.

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4.9.2 Visibility of copy test. Type a string of characters. Verify that each character typed is visible in conformance with 3.8.2.

4.9.3 Copy illumination test. Set the POWER switch to the ON position and note that the copy lamp illumination level is fully adjustable by rotating the illumination knob for conformance to 3.8.3.

4.9.4 ASCII/BAUDOT control test. Place the ASCII/BAUDOT control to the ASCII position and verify conformance to 3.8.4. Repeat with the control set to the BAUDOT position.

4.9.5 Printer function test.

4.9.5.1 Local printer function tests. Verify that local machine functions can be performed whether or not the machine is also responding to line signals in compliance with 3.8.5.1.

4.9.5.1.1 Line feed test. With power on, depress and hold the printer LINE FEED switch for at least one second. Observe that paper feeds to determine conformance with 3.8.5.1.1.

4.9.5.2 Coded BAUDOT printer functions test. Receive, generate and transmit all codes listed in Tables I, II, and III.

4.9.5.2.1 CR test. See 4.9.5.2.

4.9.5.2.2 LF test, See 4.9.5.2.

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4.9.5.2.3 LTRS test. See 4.9.5.2

4.9.5.2.4 FIGS test. See 4.9.5.2.

4.9.5.2.5 Fig. S/J switch test. Place the fig S/J switch in the S position and receive both the S and J codes to determine conformance to 3.8.5.2.5. Repeat with the switch in J position.

4.9.5.3 ASCII printer functions test,

4.9.5.3.1 Keyboard/print code test. Receive, generate and transmit all codes listed in Tables I, II, and III and verify that machine response is in accordance with requirements of Tables I, II, and III,

4.9.5.3.2 Tab code test. Receive and generate the tab code with the print position less than the largest tab setting and verify conformance to 3.8.5.3.2. Repeat with print position equal to the largest tab setting. Repeat with print position greater than the largest tab setting.

4.9.5.3.3 CR test. See 4.9.5.3.1.

4.9.5.3.4 LF test. See 4.9.5.3.1.

4.9.5.3.5 Parity error test. The following shall be performed to determine conformance to 3.8.5.3.5.

a. Even parity test. With printer PARITY switch set EVEN, receive an odd parity message to verify that the PARITY RESET lamp lights and the parity error symbol is put into memory.

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Print out message to verify. Operate the PARITY RESET switch to extinguish the lamp. Receive an even parity message to verify proper operation. Print out message to verify.

b. Odd parity test. With the printer PARITY switch set to ODD, receive an even parity message to verify that the PARITY RESET lamp lights and the parity error symbol is put into memory. Print out message to verify. Operate the PARITY RESET switch to extinquish the lamp. Receive an odd parity message to verify proper operation. Print out message to verify.

c. Parity inhibit test. With the printer PARITY switch set to INHIBIT, receive an odd parity message followed by an even parity message. No parity error should be detected. Print out the messages to verify.

d. BAUDOT operation test. With system set for BAUDOT operation, receive a message with parity switch set to ODD. The parity function should be disabled. Repeat using EVEN and INHIBIT switch settings.

e. Lamp tests. With power on, hold in the PARITY RESET switch and verify that all lamps and indicators light.

4.9.5.4 Automatic carriage return test. Using a line length of 80, type in 81 characters and verify conformance to 3.8.5.4. In the INSERT or BEFORE mode of the EDIT command, repeat using a line length of 40, and typing in 41 characters and repeat using a line length of 60, and typing in 61 characters.

4.9.5.4.1 End of line signal test. With the line length set to 80, type in 74 characters and determine conformance to 3.8.5.4.1. In the INSERT or BEFORE modes of the EDIT command, repeat with the line length set to 40 and typing in 34 characters and repeat with line length set to 60 and typing in 54 characters.

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4.9.5.5 Alarms and indicators test.

4.9.5.5.1 Audible alarm test.

4.9.5.5.1.1 Audible alarm adjustment test. With the audible alarm in a steady state condition, adjust the level and verify conformance to 3.8.5.5.1.1.

4.9.5.5.1.2 Audible alarm activation test. The following tests shall be run to verify conformance to 3.8.5.5.1.2.

a. FIG S/J codes test. With the Fig. S/J switch set to the S position (BAUDOT mode), receive the Fig. S code and verify conformance to 3.8.5.5.1.2a. Repeat with the Fig. S/J switch set to J and receive the J code.

b. Parity error test. Receive a message with at least one parity error (ASCII mode) and verify conformance to 3.8.5.5.1.2b. Activate the PARITY RESET switch and verify conformance to 3.8.5.5.1.2b.

c. BEL code test. Receive the BEL code and verify conformance to 3.8.5.5.1.2c (ASCII mode).

d. Line lamp conditions test.

(1) High level digital test. With an open line condition, a steady state alarm shall sound in conformance with 3.8.5.5.1.2d.

(2) Conditioned diphase test. With a continuous space condition a steady state alarm shall sound in conformance with 3.8.5.5.1.2d.

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(3) Low level digital test. With a continuous space condition, a steady state alarm shall sound in conformance with 3.8.5.5.1.2d.

e. Message precedence test. Receive test messages which have precedences of immediate and flash and verify conformance to 3.8.5.5.1.2e. Print the messages and verify conformance to 3.8.5.5.1.2e.

4.9.5.5.2 Power indicator test. Set illumination control to mid-level, turn power on, and verify that the copy lights are lit as required by 3.8.5.5.2.

4.9.5.5.2.1 Battery back-up indicator test. With the system in the battery back-up state, verify that the battery back-up indicator is the only indicator activated as required by 3.8.5.5.2.1.

4.9.5.5.3 Paper low indicator test. Deplete the roll paper supply to verify conformance to 3.8.5.5.3 (1-7/16 ± 1/16 inch diameter roll remains). Continue depleting the paper until none is left. The print drum and ribbon feed shall automatically stop to verify conformance to 3.8.5.5.3. Replace the supply with a new roll, enter G0 (CR) and verify that the printer can print (type a few characters) and that the PAPER LOW lamp turns off.

4.9.5.5.4 Line indicator test. Receive a test message and verify that the line lamp is "ON" during receipt of characters and is "OFF" when the line is idle as required by 3.8.5.5.4. Open the signal line (high level digital interface) and verify conformance to 3.8.5.5.4. Receive a continuous space (low level and conditioned diphase interfaces) and verify conformance to 3.8.5.5.4.

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4.9.5.5.5 Received message indicator test. Receive a test message with a precedence of less than immediate. Verify that the indicator is in a steady state condition as required by 3.8.5.5.5. Receive test messages with precedences of immediate and flash and verify conformance to 3.8.5.5.5. Print out the immediate and flash messages and verify that the light returns to a steady state condition as required by 3.8.5.5.5. Print out the less than immediate precedent message and verify that the indicator extinguishes as required by 3.8.5.5.5.

4.9.5.5.6 Transmit indicator test. Transmit a test message and verify conformance to 3.8.5.5.6.

4.9.5.5.7 Memory full warning indicator test. Fill memory to within 25% of capacity and verify conformance to 3.8.5.5.7.

4.9.5.6 Switch test. Verify by visual inspection, for conformance to 3.8.5.6.

4.9.5.6.1 PARITY switch test. Set the PARITY switch to the EVEN position and verify conformance to 3.8.5.6.1. Repeat with the switch in the ODD and INHIBIT positions.

4.9.5.6.2 Stop bits switch test. With the switch set to 1, transmit and receive a test message and verify conformance to 3.8.5.6.2. Repeat with the switch set to 2 stop bits.

4.9.5.6.3 State switch test. Place switch in the ICT position, turn power on and check mode determination printout to verify conformance to 3.8.5.6.3. Repeat with switch in the KSR and RO positions.

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4.9.5.6.4 Data inversion test. The following tests shall be performed to verify conformance to 3.8.5.6.4.

a. Set operation for noninverted transmit data. Transmit a test message.

b. Set operation for noninverted received data. Receive a noninverted test message.

c. Set operation for inverted received data. Receive a noninverted test message.

d. Set operation for inverted received data. Receive an inverted test message.

4.9.5.6.5 Self test switch test. Activate the self test switch and verify conformance to 3.8.5.6.5.

4.9.5.6.6 ABORT switch test. Activate the ABORT switch during operation in the ICT mode for each of the following conditions and verify conformance to 3.8.5.6.6.

a. Message transmission

b. Message reception

c. Message printout

4.9.5.6.7 Audio alarm reset test. With a steady alarm condition present, activate the AUDIO ALARM RESET switch and verify conformance to 3.8.5.6.7.

4.9.5.6.8 External power switch test. Activate the external power switch and verify conformance to 3.8.5.6.8.

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4.9.6 System states.

4.9.6.1 Off state test. N/A.

4.9.6.2 Cold start state test. Turn power on and check printout to verify conformance to 3.8.6.2.

4.9.6.3. Operation validation/state determination/state test. The following tests shall be performed to verify conformance to 3.8.6.3.

a. Test 1. Set all switches to valid setting. Turn power "on" and check printout.

(1) Set: BAUDOT, 45.5 Baud, Rec/Xmit Mode: High Level, Non-Inverted, KSR State, 2 Stop Bits, Fig. S, NRZ and +INT Clock.

(2) Repeat once using different valid switch setting combinations.

b. Test 2. Set mode switch to ASCII, Baud Rate switch to 50 and the remaining switches to valid settings. Check error printout.

c. Test 3. Set mode switch to BAUDOT, Baud Rate switch to 150, and the remaining switches to valid settings. Check error printout.

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d. Test 4. Set mode switch to BAUDOT, signal switch to Conditioned Diphase, and the remaining switches to valid settings. Check error printout.

e. Test 5. Set mode switch to ASCII, signal switch to Conditioned Diphase, clock switch to external, 150 Baud, 1 stop bit, check error printout.

4.9.6.4 ICT state test. Set the state switch to the ICT position, and perform the following tests to verify conformance to 3.8.6.4.

a. Apply prime power and verify entrance into the System Command mode.

b. Enter the following commands and verify entrance into the proper modes.

(1) EDIT

(2) PARAMETER

c. Disconnect prime power with battery back-up present and verify memory save capability.

d. Activate the self test switch and verify appropriate printout.

4.9.6.5 RO state test. Set the switch to the RO position, and perform the following test to verify conformance to 3.8.6.5.

a. With message memory physically missing, receive a test message at 600 Baud.

b. With message memory present, recerive a test message at 150 Baud.

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c. With message memory present, receive a test message at 1200 Baud. The length of the message should be less than message storage capacity.

d. Receive a test message which contains a communications envelope, end of line sequence and redundant carriage returns.

4.9.6.6 KSR state test. Set the state switch to the KSR position. and perform the following tests to verify conformance to 3.8.6.6.

a. Operate the DLC, DLL and REV keys.

b. Transmit a one line test message.

c. Receive a test message.

d. Receive a test message which contins a communications envelope, end of line sequence and redundant carriage returns.

4.9.6.7 Battery Back-up State test. With equipment operating on prime power and back-up battery present, remove prime power and verify conformance to 3.8.6.7. Return prime power and verify conformance to 3.8.6.7.

4.9.6.8 Self test state test.

a. Activate the self test switch with the equipment operating on prime power to verify conformance to 3.8.6.8.

b. Activate the self test switch with the equipment operating on battery back-up and verify conformance to 3.8.6.8.

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c. A series of tests shall be performed, in which faults are introduced into the equipment and correctly recognized by the self-test function.

4.9.7 Message processing test. Verify that the basic editing functions are available for all keyboard inputs as required by 3.8.7.

4.9.7.1 Message storage test.

4.9.7.1.1 Memory full test.

4.9.7.1.1.1 Operator input test. With 464 characters unallocated, input an 80 character line via keyboard and verify conformance to 3.8.7.1.1.1.

4.9.7.1.1.2 Incoming message. With 464 characters unallocated (received messages are present in memory), receive a message of greater than 80 characters and verify conformance to 3.8.7.1.1.2. Repeat with no received message present in memory.

4.9.7.2 Message receipt tests. Receive multiple messages, and verify that they are assigned consecutive numbers and that the MSG RCVD lamp is activated to verify conformance to 3.8.7.2.

4.9.7.2.1 Receive Message Process test. The following tests shall be conducted using two AN/UGC-74A(V)'s back-to-back to verify conformance to 3.9.7.2.1.

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a. Null the total respective transmit envelope of unit 1 and receive envelope of unit 2. Transmit a test message from unit 1 to unit 2 and terminate transmission before completion and assure that no characters are received for 10 seconds. Verify that the partial message received was saved.

b. Set the total respective tenvelope of unit 1 and the renvelope of unit 2 to the identical non-null sequence. Transmit a test message from unit 1 to unit 2 and terminate transmission before completion and assure that no characters are received for 10 seconds. Verify that the partial message received was not saved.

c. Set the total respective tenvelope of unit 1 and the renvelope of unit 2 to the identical non-null-sequence. Transmit a test message from unit 1 to unit 2 to completion. Verify that the message has been saved.

d. Null the front of envelope of units 1 (TENV) and 2 (RENV). Set the close of envelopes to be an identical non-null sequence. Transmit a test message from unit 1 to unit 2 and terminate transmission before completion and assure that no characters are received for 10 seconds. Verify that the partial message has not been saved in memory.

e. Set the front of tenvelope of unit 1 and the front of renvelope of unit 2 to the identical non-null sequence (close of envelopes are null). Transmit two different test messages within 10 seconds and verify that the first message is saved.

f. Repeat c with units 1 and 2 having identical end of line sequences which are not a single CR. Verify that the EOL sequence has been replaced by a single CR.

4.9.7.3 Transmit message process test. With transmit envelope and end of line sequence set, transmit a test message and verify conformance to 3.8.7.3. Transmit a test message, and terminate before completion by activating the ABORT switch and verify conformance to 3.8.7.3.

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4.9.7.4 System Commands test. All software commands shall be input using two separate forms, (1) Full Text, (2) the shortest truncated form with at least one leading space, at least one trailing space and at least one interleaved space, if applicable to verify conformance to 3.8.7.4 (No special test is necessary, as this test can be combined with other tests),

4.9.7.4.1 PARAMETER Command test. Enter "PARAMETER" via keyboard and verify that PARAMETER Sub-commands are made available to the operator as required by 3.8.7.4.1.

4.9.7.4.1.1 LINE Sub-command test.

a. Verify line length is set to 80 on power up as required by 3.8.7.4.1.1 The following shall be entered and conformance to 3.8.7.4.1.1 determined.

b. LINE (60), type in 61 printing characters (Insert or Before in EDIT Command)

c. LINE (40), type in 41 printing characters (Insert or Before in EDIT Command)

d. LINE (80), type in 81 printing characters

e. LINE (35)

f. LINE (90)

4.9.7.4.1.2 SETAB Sub-command test. The following shall be entered to verify conformance to 3.8.7.4.1.2.

a. SETAB (Number(s) less than or equal to line length)

b. SETAB (Number(s) greater than line length)

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c. SETAB (Previously set value)

d. SETAB(13 or more values)

4.9.7.4.1.3 CLRTAB Sub-command test. The following shall be entered to verify conformance to 3.8.7.4.1.3.

a. CLRTAB (Set Tab Value)

b. CLRTAB (*)

c. CLRTAB (Not Set Tab Value)

d. CLRTAB (No Tab Set)

4.9.7.4.1.4 DISPLAY Sub-command test. With tabs set enter the DISPLAY Sub-command and verify conformance to 3.8.7.4.1.4. Repeat with no tabs set.

4.9.7.4.1.5 CAPIT Sub-command test. With CAPIT option set transmit lower case alpha characters (keyboard unshifted) and verify conformance to 3.8.7.4.1.5. Repeat with CAPIT not set. Verify that CAPIT is active upon power up as required by 3.8.7.4.1.5.

4.9.7.4.1.6 REMIND Sub-command test. With REMIND option set, enter TRANSMIT (CR) and verify conformance to 3.8.7.4.1.6. Upon power up, verify that REMIND is inactive as required by 3.8.7.4.1.6.

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4.9.7.4.1.7 EOL Sub-command test. The following shall be entered to verify conformance to 3.8.7.4.1.7.

a. EOL (8 or less characters)

b. EOL (more than 8 characters)

c. EOL (improper entry of less than 8 characters)

Upon power up, verify that EOL sequence is CR CR LF. Enter EOL Command during message transmission.

4.9.7.4.1.8 LF Sub-command test. The following shall be entered to verify conformance to 3.8.7.4.1.8.

a. LF (1)

b. LF (2)

c. LF (any number except 1 or 2)

Verify upon power-up that LF is set to 1

4.9.7.4.1.9 SPACE Sub-command test.

a. Verify that upon power-up the space option is "OFF". The following shall be entered to verify conformance to 3.8.7.4.1.9.

b. SPACE (ON), Compose and transmit a test message containing tabs

c. SPACE (OFF), Compose and transmit a test message containing tabs

d. SPACE (Any character sequence except "ON" or "OFF")

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4.9.7.4.1.10 PAPER Sub-command test. Perform the following tests to verify conformance to 3.8.7.4.1.10. Enter a test message into memory.

a. Enter PAPER.

b. With print drum and ribbon feed off, verify that the LF switch is active.

c. With print drum and ribbon feed off, enter GO (CR). Verify return to the PARAMETER Sub-command mode level.

d. With print drum and ribbon feed off, enter in any sequence of characters except GO (CR).

e. Deplete paper supply completely.

4.9.7.4.1.11 RENVELOPE Sub-command test. Perform the following tests to verify conformance to 3.8.7.4.1.11.

a. Enter RENVELOPE Sub-command while a test message is being received.

b. Verify that upon power-up with the system in BAUDOT mode, the RENVELOPE is NULL.

c. Verify that upon power-up with the system in ASCII mode, the RENVELOPE is VZCZC:NNNN.

The following shall be entered to verify conformance to 3.8.7.4.1.11.

d. RENVELOPE (0 to 8 characters of start sequence:
0 to 16 characters of stop sequence)

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e. RENVELOPE (0 to 8 characters of start sequence:
17 or more characters of stop sequence)

f. RENVELOPE (9 or more characters of start sequence:
0 to 16 characters of stop sequence)

g. RENVELOPE (Improper character group)

4.9.7.4.1.12 TENVELOPE Sub-command test. Perform the following tests to verify conformance to 3.8.7.4.1.12.

a. Enter TENVELOPE Sub-command while a test message is being transmitted.

b. Verify that upon power-up with the system in BAUDOT mode, the TENVELOPE is NULL.

c. Verify that upon power-up with the system in ASCII mode, the TENELOPE is:

VZCZC:NNNN $\overbrace{\text{DEL- - - - DEL}}^{12X}$

The following shall be entered to verify conformance to 3.8.7.4.1.12

d. TENVELOPE (0 to 8 characters of start sequence:
0 to 16 characters of stop sequence)

e. TENVELOPE (0 to 8 characters of start sequence:
17 or more characters of stop sequence)

f. TENVELOPE (9 or more characters of start sequence:
0 to 16 characters of stop sequence)

g. TENVELOPE (Improper character group)

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4.9.7.4.1.13 EXIT Sub-command test. Enter the EXIT Sub-command and verify entry into the "System Command mode" from EDIT and PARAMETER modes.

4.9.7.4.2 EDIT Command test. The following shall be entered and conformance to 3.8.7.4.2 verified.

a. EDIT (NULL)

b. EDIT (Message No. in memory)

c. EDIT (Message No. not in memory)

Enter EDIT command via keyboard, and verify that EDIT Sub-commands are made available to the operator as required by 4.9.7.4.2.

4.9.7.4.2.1 TOP Sub-command test. Enter a test message. Enter the TOP Sub-command and verify conformance to 3.8.7.4.2.1. With a NULL message, enter TOP Sub-command and verify error printout.

4.9.7.4.2.2 BOTTOM Sub-command test. Enter a test message. Enter the BOTTOM Sub-command and verify conformance to 3.8.7.4.2.2. With a NULL message enter the BOTTOM Sub-command and verify error printout.

4.9.7.4.2.3 UP Sub-command test. Enter a test message. The following shall be entered and conformance to 3.8.7.4.2.3 verified.

a. UP (specified no. less than no. of lines from current line pointer position to first line of message).

b. UP (null)

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c. UP (*)

d. UP (100 or greater)

e. UP (2 non-numeric characters)

f. UP (specified no. greater than no. of lines from current line pointer position to first line of message.

Enter a NULL test message. Enter UP (specified number) and verify conformance to 3.8.7.4.2.3.

4.9.7.4.2.4 DOWN Sub-command test. Enter a test message. The following shall be entered and conformance to 3.8.7.4.2.4 verified.

a. DOWN (specified number less than the number of lines from current line position to last line of message)

b. DOWN (NULL)

c. DOWN (*)

d. DOWN (100 or greater)

e. DOWN (2 non-numeric characters)

f. DOWN (specified number greater than the number of lines from the current line position to last line of message)

Enter a NULL message. Enter DOWN (specified number) and verify conformance to 3.8.7.4.2.4.

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4.9.7.4.2.5 BEFORE Sub-command test. Enter a test message. Enter a block of text before a designated line, terminate command by depressing HLT key, and verify that the requirements of 3.8.7.4.2.5 are met by printing out entire message. Enter a block of text when sufficient memory is not available for allocation and verify conformance to 3.8.7.4.2.5.

4.9.7.4.2.6 INSERT Sub-command test. Enter a test message. Enter a block of text after a designated line, terminate command by depressing HLT key, and verify that the requirements of 3.8.7.4.2.6 are met by printing out entire message. Enter a block of text when sufficient memory is not available for allocation and verify conformance to 3.8.7.4.2.6.

4.9.7.4.2.7 DELETE Sub-command test. Enter a test message. The following shall be entered and conformance to 3.8.7.4.2.7 verified.

a. DELETE (specified number of lines)

b. DELETE (NULL)

c. DELETE (*)

d. DELETE (specified number of lines greater than number of lines remaining)

e. DELETE (specified number of lines equal to total message length with the line pointer at first line)

f. DELETE (100 or greater)

g. DELETE (2 non-numeric characters)

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4.9.7.4.2.8 FIND Sub-command test. Enter a test message. The following shall be entered and conformance to 3.8.7.4.2.8 verified.

a. FIND (#specified phrase of 20 characters or less which is contained in the test message #)

b. FIND (#specified phrase of 21 characters which is contained in the test message#)

c. FIND (#NULL#)

d. FIND (#specified phrase containing a CR#)

e. FIND (#specified phrase not contained in the test message#)

4.9.7.4.2.9 LIST Sub-command. Enter a test message. The following shall be entered and conformance to 3.8.7.4.2.9 verified.

a. LIST (Specified number of lines less than the number of lines from the current line to the end of the test message)

b. LIST (NULL)

c. LIST (*)

d. LIST (100 or greater)

e. LIST (2 non-numeric characters)

f. LIST (specified number of lines greater than the number of lines from the current line to the end of the message)

Enter a NULL message. Enter LIST (specified number of lines) and verify conformance to 3.8.7.4.2.9. Depress HLT key during message listing and verify conformance to 3.8.7.4.2.9.

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4.9.7.4.2.10 REPLACE Sub-command test. Enter a test message. The following shall be entered and conformance to 3.8.7.4.2.10 verified:

a. REPLACE (#specified phrase of 1 to 20 characters contained in one line of the test message #specified phrase of 0 to 40 characters#)

b. REPLACE (#specified phrase of 1 to 20 characters not contained in the test message##)

c. REPLACE (#specified phrase of 0 characters##)

d. REPLACE (#specified phrase of 21 characters contained in the test message##)

e. REPLACE (#specified phrase of 1 to 20 characters contained in one line of the test message #specified phrase of 41 characters#)

4.9.7.4.2.11 HEADER Sub-command test. With line length set to 69 or less, enter a valid header which contains the following and verify conformance to 3.8.7.4.2.11.

a. Data with and without leading and trailing spaces.

b. Numbers with and without leading zeros.

c. Entries with improper number of characters.

d. Entries with improper types of characters.

With line length set to 70, enter HEADER Sub-command and verify message printout for conformance to 3.8.7.4.2.11.

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4.9.7.4.2.12 APPEND Sub-command test. Enter a test message. The following shall be entered and conformance to 3.8.7.4.2.12 verified.

a. APPEND (message no. in memory (message contains no parity errors))

b. APPEND (message no. in memory (message contains parity errors))

c. APPEND (message not in memory)

d. APPEND (message no. of currently edited message)

Enter the APPEND command when insufficient memory is available for allocation and verify conformance to 3.8.7.4.2.12.

4.9.7.4.2.13 EXIT Sub-command test. Enter the EXIT Sub-command and verify entry into System Command mode from EDIT and PARAMETER modes (see 3.8.7.4.2.13).

4.9.7.4.3 PRINT Command test. Enter a test message. The following shall be entered and conformance to 3.8.7.4.3 verified.

a. PRINT (message number in memory)

b. PRINT (message number not in memory)

c. PRINT (*)

Enter the PRINT command, halt printing, and verify conformance to 3.8.7.4.3.

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4.9.7.4.4 TRANSMIT Command test. The following shall be entered and conformance to 3.8.7.4.4 verified.

a. TRANSMIT (message number in memory (message is not NULL))

b. TRANSMIT (message number not in memory)

c. TRANSMIT (message number in memory with NNN) (Characters 4,5, and 6 in line 1 are numeric)

d. Repeat c with characters on line one, not numeric

e. TRANSMIT (receive message number in memory)

Depress the HLT key during transmission and verify conformance to 3.8.7.4.4. Enter the TRANSMIT command during message transmission and verify conformance to 3.8.7.4.4.

4.9.7.4.5 REMOVE Command test. The following shall be entered and conformance to 3.8.7.4.5 verified.

a. REMOVE (received, printed message number in memory)

b. REMOVE (message number not in memory)

c. REMOVE (received, non-printed, message number in memory)

d. REMOVE (composed, transmitted, message number in memory)

e. REMOVE (composed, message in transmission, message number in memory)

f. REMOVE (composed, non-transmitted, message number in memory)

g. Repeat f.

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4.9.7.4.6 JUSTIFY Command test. The following shall be entered and conformance to 3.8.7.4.6 verified.

a. JUSTIFY (specified message in system)

b. JUSTIFY (specified message not in system)

4.9.7.4.7 QUERY Command test. With the system set for ASCII operation, enter the QUERY Command and check printout to verify conformance to 3.8.7.4.7. Repeat with system set for BAUDOT operation.

4.9.7.4.8 STATUS Command test. The following shall be entered and conformance to 3.8.7.4.8 verified.

a. STATUS (specified message(s) in system)

b. STATUS (specified message not in system)

c. STATUS (*)

d. STATUS (NULL)

4.9.7.4.9 TTY Command test. Enter the TTY command with the system operating with the following conditions present and verify conformance to 3.8.7.4.9.

a. Composed message being transmitted.

b. Previous TTY command message being transmitted.

c. Transmitter is free.

d. No text.

4.9.7.5 Full duplex operation test. The equipment under test shall have a test message in memory. A message shall be

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transmitted to the equipment under test from a test message generator and at the same time the test message in memory shall be transmitted to a receive printer. The simultaneous transmit-receive process shall not cause any errors or loss of characters to verify compliance with 3.8.7.5.

4.9.8 Performance requirements tests.

4.9.8.1 Response times test. Verify compliance to 3.8.8.1.

4.9.8.2 Storage capacity test. Remove the message storage memory block. Set the equipment for RO mode of operation. Receive a test message. The message shall be received and printed properly without loss or error to comply with 3.8.8.2.

4.9.8.2.1 Memory retention. Connect an external power source capable of sustaining message storage. With the equipment operating and the message storage filled with copies of a test message, disconnect prime power. Reconnect prime power and print out the message storage contents to verify that it has not been lost or altered due to the power outage to verify conformance to 3.8.8.2.1.

4.9.8.3 Printing speed test. Print a test message of at least 20 eighty-character lines and verify conformance to 3.8.8.3.

4.9.8.4 Print quality test. Print a test message of at least one line of 80 letter E's and verify comformance to 3.8.8.4. by observing that the top and bottom of letter is visible.

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4.9.8.5 Automatic time out test. When set in the KSR mode and prime power on, print a character, and wait a minimum of one minute, fifteen seconds, to verify automatic time out as required by 3.8.8.5. After the time out has occurred, enter a string of characters and verify that none are lost to verify conformance to 3.8.8.5. Repeat above, receive a string of characters.

4.9.9 Operating prompting test.

4.9.9.1 Error/information list test. Print out all information and error messages and verify conformance to 3.8.9.1. (No special test necessary, may be combined with other tests).

4.9.10 Transfer switch test. Operate the transfer switch and verify that closure is made in accordance with 3.8.10.

4.10 AN/UGC-74A(V) signal characteristics and electrical requirements test.

4.10.1 Operating modes test.

4.10.1.1 ASCII code interface test. Transmit and receive an ASCII coded test message to verify proper operation using each of the following interfaces to determine compliance to 3.9.1.1.

a. Balanced low level digital interface using internal clock.

b. Balanced low level digital interface using external clock.

c. High level digital interface using internal clock.

d. High level digital interface using external clock.

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4.10.1.1.1 Signal speed test (ASCII). Transmit and receive an ASCII coded test message to verify proper operation at each of the following internally clocked signal speeds to determine compliance with 3.9.1.1.1.

a. ASCII, 10 unit, 300, 1200 bps (balanced low level)

b. ASCII, 10 unit, 75 bps (high level)

c. ASCII, 11 unit, 600 bps (balanced low level)

d. ASCII, 11 unit, 150 bps (high level)

4.10.1.1.2 External clock mode test. Transmit and receive an ASCII coded test message to verify proper operation at each of the following externally clocked signal speeds to determine compliance with 3.9.1.1.2.

a. ASCII, 10 unit, 75 bps (high level)

b. ASCII, 11 unit, 1200 bps (balanced low level)

4.10.1.2 BAUDOT code interface test. Transmit and receive a BAUDOT coded test message to verifty proper operation using each of the following interfaces to determine compliance with 3.9.1.2.

a. Balanced low level digital interface using internal clock.

b. Balanced low level digital interface using external clock.

c. High level digital interface using internal clock.

d. High level digital interface using external clock.

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4.10.1.2.1 Signal speed test (BAUDOT). Transmit and receive a BAUDOT coded test message to verify proper operation at each of the following internally and externally clocked signal speeds to determine compliance with 3.9.1.2.1.

a. BAUDOT, 7 unit, 45.5 bps

b. BAUDOT, 8 unit, 50 bps

c. BAUDOT, 8 unit, 75 bps

4.10.1.3 Speed accuracy test.

a. Operate the equipment in the BAUDOT mode and transmit repeated Y's at each of the following signal speeds and verify conformance to 3.9.1.3.

(1) 45.5 bps

(2) 75 bps

b. Operate the equipment in the ASCII mode and transmit repeated U's at each of the following signal speeds and verify conformance to 3.9.1.3.

(1) 150 bps

(2) 1200 bps

4.10.2 Clock INT/EXT/KG-30 switch test. With the clock switch set for internal operation, verify that the unit operates at the signal speed selected by the BAUD RATE switch to comply with 3.9.2. Repeat with the clock switch set to the external position. With the clock switch set for KG-30 operation, verify that the equipment can accept external clocks for the transmit and receive function to comply with 3.9.2.

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4.10.2.1 Internal clock mode-transmit clock test. With the equipment set in the internal clock mode, verify that the clock is made available to other units at the connector. Verify that the clock rate corresponds to the BAUD RATE switch setting. With the system set for non-inverted clock, verify the transmit data corresponds to the positive going slope of the transmit clock as required by 3.9.2.1. Repeat using inverted clock.

4.10.2.2 External clock/date relationship test.

4.10.2.2.1 External clock/data transmitted relationship test. Arrange the printer for external clock, not inverted, 1200 bps. Transmit the letter "U" (ASCII) and check that the relationship between the clock signal and data signal wave shapes conform to those of Figure 1 to determine conformance to 3.9.2.2.1. Repeat with inverted clock.

4.10.2.2.2 External clock/received data relationship test. Receive the letter "U" and check that the relationship between the clock signal and data (ASCII) signal waveforms conform to Figure 1 to determine compliance to 3.9.2.2.2.

4.10.2.3 Data transmission test. Transmit the letter "Y" (BAUDOT) and check that the relationship between the internal clock and data waveforms conform to those of Figure 1 (+4.5 percent) to determine compliance with 3.9.2.3. Repeat for external clock.

4.10.3 Detailed requirements for digital interface test. The equipment shall be tested for compliance with 3.9.3. The test procedure shall be in accordance with MIL-STD-188-114, dated 24 March 1976, Section 5.1 (exclusions and changes in 3.9.3 shall be applicable.

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4.10.3.1 Signals test. Transmitter waveshapes shall be observed to asssure that the NRZ and conditioned diphase signals are provided as required by 3.9.3.1 and 3.9.3.1.1.

4.10.3.1.1 Conditioned diphase test. See 4.10.3.1.

4.10.3.2 High level digital signal interface test.

4.10.3.2.1 Idle transmit condition test. Verify that the idle transmit condition conforms to 3.9.3.2.1.

4.10.3.2.2 Transmitter source impedance test. Verify the source impedance of the high level (60mA) transmit interface does not exceed 150 ohms to determine conformance to 3.9.3.2.2.

4.10.3.2.3 Receive interface (180 V operation) test. With the signal circuit set for 60 mA, verify that the input sensitivity is in the 4 mA operating differential region to determine conformance to 3.9.3.2.3. Repeat with the signal circuit set for 20 mA.

4.10.3.3 Normal input keying (NIK) interface test. Operate the equipment in ASCII, 11 unit code at 150 bps with external clock. KW-7 input capacitance shall be adjusted to its maximum value, Verify that the equipment conforms to the operating and distortion requirements of 3.9.3.3.

4.10.4 Signal distortion test.

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4.10.4.1 Receive equipment test. Applying the test criteria of 4.10.4.1.1 (using less than 50 percent of indicated maximum) and 4.10.4.1.2 (50 percent of indicated maximum) simultaneously, verify conformance to 3.9.4.1.

4.10.4.1.1 Digital distortion test. Set the distortion test set for 42 percent swinging distortion. Set the receive unit under test for internal clock. Send a test message and check the printed copy for accuracy to determine conformance to 3.9.4.1.1. Return the distortion test set to zero distortion. Repeat using conditioned diphase data and 21 percent distortion.

4.10.4.1.2 Spurious pulses test.

a. Set the spurious pulse test set for spurious signals having a duration of 24 percent of the bit rate and using external clock.

b. Send a test message and check that no errors occur in the printed copy to determine conformance to 3.9.4.1.2.

c. Set the spurious pulse test set for spurious signals separated by 9 percent of the bit rate and consecutive pulses having a combined pulse width of 24 percent of the bit rate.

d. Send a test message and check that no errors occur in the printed copy to determine conformance to 3.9.4.1.2.

4.10.4.2. Transmit equipment test. Verify that the transmitted output data signals have less than two percent bias or end distortion at 75 bps $\pm$0.1 percent to determine conformance to 3.9.4.2. Repeat for 45.5 bps.

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4.10.5 Isolation of chasis ground, signal and clock lines test. Connect each cable, one at a time, to the unit under test, and with an ohmmeter measure (power off) between the interface external ground terminal and the ground pin of the free end of the cable to determine conformance to 3.9.5 (third line grounded to cable shield). Repeat the measurement between each pin and the ground pin to determine conformance to 3.9.6 (lines isolated from chassis ground). Repeat with ohmmeter leads reversed. Minimum resistance shall be 100K ohms.

4.10.6 System interface test. Receive and transmit a test message using the following configuration as required by 3.9.6.

EQUIPMENT COMPATIBILITY TEST

EQUIPMENT TESTED	BAUD RATE		INTERFACE
	BAUDOT	ASCII	
KW-7 (NIK Interface	75	150	1-1/2V, 70 micro-amp transmit and 48V, 20 mA receive

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4.10.7 Power test.

4.10.7.1 Power consumption test. Place the equipment under test in the maximum power mode (illumination controls on maximum and keyboard sending to printer). Measure the power consumption when operating the equipment from each of the following power sources as required by 3.9.7.1.

a. 26 VDC

b. 115 VAC, 60 Hz

c. 230 VAC, 380 Hz

d. 132 VAC, 47.5 Hz

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4.10.7.2. Sources test. Receive and transmit a test message using each of the following power sources to verify conformance to 3.9.7.2.

a. 22 VDC

b. 97.5 VAC, 400 Hz

c. 195 VAC, 380 Hz

d. 265 VAC, 60 Hz

e. 26 VDC

f. 115 VAC, 60 Hz

4.10.7.3 Over voltage and transients test. The equipment shall be stabilized by operating it for an hour before performing any test. All tests required to verify compliance with 3.9.7.3.1 and 3.9.7.3.2 shall be performed at 77 degrees ± 5 degrees F.

4.10.7.3.1 AC overload test.

a. Subject the equipment, while operating, to an input voltage of 158.7 VAC, 60 Hz for a period of five minutes.

b. During the over voltage test no damage to the equipment shall occur.

c. Data loss or degradation shall not occur as required by 3.9.7.3.1.

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4.10.7.3.2 DC over voltage and transient voltage protection test.

a. Subject the equipment to a continuous input voltage of 36 VDC for a period of one hour while operating and verify conformance to 3.9.7.3.2a.

b. Simulated voltage spikes shall be applied to the equipment while it is operating at nominal voltage. The test spike shall have an amplitude of 250 volts, a risetime not exceeding 50 nanoseconds, a frequency of oscillation greater than 100 kHz and less than 500 kHz and an energy content of not less than 15 millijoules. Tests shall be carried out with both polarities of spike voltage. A minimum of fifty 250-volt spikes of each polarity shall be applied at one second intervals. The voltage spikes so imposed shall not cause any damage to the equipment to verify conformance to 3.9.7.3.2.b.

c. Subject equipment, while operating at minimum voltage, to 15 VDC (transient) for a period of at least 75 ms, at the repetition rate of 10 pulses per minute for one hour and verify conformance to 3.9.7.3.2c.

4.10.7.4 Isolation of chassis ground power line test. Connect each cable, one at a time, to the unit under test, and with an ohmmeter measure (power off) between the interface external ground terminal and the ground pin of the free end of the cable to determine conformance to 3.9.7.4 (a separate line shall carry chassis ground). Repeat the measurement between each power line pin and the ground pin to determine conformance to 3.9.7.4 (no power source shall be grounded to the chassis). Repeat with ohmmeter leads reversed. Minimum resistance shall be 100 K ohms.

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4.11 Keyboard Transmitter TT-745/UGC-74A(V) requirements test.

4.11.1 Operating modes test. N/A

4.11.2 Keyboard arrangement test. Verify by visual inspection.

4.11.3 Two key rollover test. Proceed as follows to determine conformance to 3.10.3.

a. Depress and hold down any printing character key. See that the character is printed.

b. Depress and hold down a second printing character key. See that the character is not printed.

c. Release the first key depressed. See that the character from the second key depressed is printed.

d. Release the second key depressed. Observe that no other printing occurs.

4.11.3.1 Key pressure test. Test at random 10 percent of all keys in addition to the space bar to determine that pressure required for operation is from 3 through 6 ounces to verify conformance to 3.10.3.1.

4.11.4 Keyboard tests.

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4.11.4.1 Lower case characters test. Enter and transmit each of the lower case characters on the keytops with the keyboard in the normal state (neither the shift nor control key operated) to determine conformance to 3.10.4.1.

4.11.4.2 Shift key test. Repeat the procedure of 4.11.4.1 with the shift key depressed and determine conformance to 3.10.4.2.

4.11.4.3 Shift lock key test. Lock the SHIFT key down by depressing the LOC key. Verify that depression of the SHIFT key returns the keyboard to its normal state (LOC and SHIFT keys unlocked) as required by 3.10.4.3. Verify indication of keyboard lock in shift position.

4.11.4.4 Machine functions test. With the keyboard in each of its three positions (normal, shift, and control), type and transmit CR, LF, BS, DEL, TAB and SPACE to determine conformance to 3.10.4.4.

4.11.4.5 Control codes test. With the control key depressed, type each of the control characters (shown on the left side of each key) and verify the transmitted codes to determine conformance to 3.10.4.5. Repeat with the SHIFT key depressed.

4.11.4.6 Control keys test. With the control key depressed, type each of the keys without control characters and verify conformance with 3.10.4.6.

4.11.4.7 Repeat key test. Depress a printing character key followed by the RPT key and verify conformance to 3.10.4.7.

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4.11.4.8 TAB key test. With tabs set, compose a test message utilizing the TAB key at least two times. Transmit the test message, with the SPACE option set, and verify conformance to 3.10.4.8. Repeat with SPACE option not set.

4.11.4.9 Message processing keyboard test.

a. With the equipment operating in the BAUDOT mode, depress the DEL and BS keys and verify conformance to 3.10.4.9.

b. DLC key test - Enter a string of characters. Depress the DLC key and verify conformance to 3.10.4.9. Repeat using a null line.

c. DLL key test. Enter a string of characters and verify conformance to 3.10.4.9. Repeat using a null line.

d. REV key test. Enter a string of characters. Delete the last character. Depress the REV key and verify conformance to 3.10.4.9. Depress the REV key with a null line present and verify conformance to 3.10.4.9.

e. HALT key test. With the equipment operating under the following conditions, depress the "HALT" key and check the system reponse to verify conformance to 3.10.4.9.

(1) Printing

(2) INSERT Sub-command

(3) BEFORE Sub-command

(4) HEADER Prompting Sequence

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4.12 TEMPEST. The equipment shall be subjected to the First Article and Group D tests specified in MIL-T-49261. The sampling rate shall be one unit tested per 600 produced under the contract. Samples shall be selected and tested from the initial production lot and at the beginning of each successive lot.

4.13 Electromagnetic compatibility. The equipment shall be tested for compliance with the requirements of 3.13. Test procedures shall be in accordance with the test methods of MIL-STD-462 Notice 3 as implemented by a contractor prepared, Government approved test plan. The subsequent Group D sampling shall be one unit from each 600 produced under the contract.

4.14 Nuclear hardness validation. The order of nuclear tests shall be as follows: first, EMP; second, gamma dose-rate; third, gamma total dose; and fourth, neutron fluence. The tests shall be in accordance with 4.14 .1 through 4.14.3 on one first article unit. All items delivered after first article testing will require certification from the contractor that all active and passive components are identical to those delivered with the first article equipment. The contractor shall also certify that there has been no change in material and thickness of the combination cases, and that workmanship of the cases for EMP survival is at least equal to that of the acceptable first article equipment.

4.14.1 Tests and analyses. Tests and analyses shall be performed on the equipment in its intended operating configuration to assure the system nuclear survivability as specified in 3.14. Specific areas of consideration are discussed in paragraphs 4.14.2 thru 4.14.3 inclusive. As required in the contract.

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Calibration certification of the test facilities to be used are required by the COTR, 45 days prior to nuclear test. Verification of the wave shape of the test simulator is required. Calibration and wave shape verification shall be supplied to the COTR by the contractor.

4.14.2 Electromagnetic pulse. The equipment shall be subjected to the EMP environments as required in 3.14. The equipment shall be placed within a ground-effect electromagnetic pulse facility in the following orientation:

a. Connecting cables extended parallel to the electric field and perpendicular to the magnetic field; all individual units must be in the test volume but widely separated.

The equipment shall be exposed at threat level in the above orientation and in each operating mode of the equipment. The equipment shall meet the operational requirements of Section 3 upon completion of threat level tests. Where necessary, test data to satisfy the EMP requirements may be obtained by extrapolation of diagnostic test levels to threat level and injection pulsing at the calculated level.

4.14.3 Initial nuclear radiation. For the purpose of design, but not validation testing, electronic pieceparts and circuits for which nuclear test data are available or for which an acceptable level of confidence has been established through mathematical analysis need not be subjected to further nuclear design testing. Acceptance of these electronic pieceparts, circuits and mathematical analysis will be determined by the procuring agency from data supplied to the Government by the contractor.

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4.14.3.1 Neutron fluence. The equipment shall be exposed in such a manner that all electronic pieceparts and circuits receive the Neutron Fluence required in 3.14. The equipment shall then perform as specified in Section 3.

4.14.3.2 Gamma total dose. The equipment shall be exposed in such a manner that all electronic pieceparts and circuits receive the Gamma Total Dose required in 3.14. The equipment shall be energized during exposure and shall then perform as specified in Section 3. Except for fiber optics and n-channel metal-oxide semiconductors (NMOS), verification of device hardness against permanent damage from gamma ray dose may be achieved by the gamma output which normally accompanies the neutron fluence. Gamma dose hardness of fiber optics and NMOS devices must be verified by Cobalt-60 irradiation in which the total dose specified is delivered in a steady-state exposure on the order of 20 seconds duration.

4.14.3.3 Peak gamma dose rate. The equipment shall be exposed in such a manner that all electronic pieceparts and circuits receive the Peak Gamma Dose Rate required in 3.14. The equipment shall be energized during exposure and shall then recover and perform as specified in 3.14.

4.15 Environmental. The following tests shall be performed on the equipment (See 3.15).

4.15.1 Temperature.

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4.15.1.1 High temperature. The equipment shall be subjected to test Method 501.1, Procedure II, of MIL-STD-810. During Step 7, the internal chamber temperature shall be adjusted to +155 degrees F (+68degrees C) and the verification test in accordance with 4.19 shall be performed (See 3.15.1).

4.15.1.2 Low temperature. The equipment shall be subjected to test Method 502.1, Procedure I of MIL-STD-810. The storage temperature in step 2 shall be -65 degrees F (-54 degrees C) and shall be maintained for a period of not less than 2 hours following stabilization of the test item. Low operating temperature (step 4) shall be -25 degrees F (-31.7 degrees C) and the verification test in accordance with 4.19 shall be performed (See 3.15.1). A special low temperature ribbon, number 3221-17605 with number 2563 ink as manufactured by Columbia Ribbon and Carbon, Blytheville, Arkansas or equivalent may be used during the low temperature test (see 3.15.1).

4.15.2 Humidity. The equipment with case open shall be subjected to the test Method 507.1, Procedure II, of MIL-STD-810. The equipment shall be operated once in every 24 hour period (4.19) and during the last five hours of the final cycle of Step 6 and as otherwise specified in Procedure II. This test shall be performed with the required roll of paper installed. However, if the condition of the paper is such that it will interfere with proper operation of the equipment, then changing the roll of paper is permitted. After the humidity test, the verification test in accordance with 4.19 shall be performed (See 3.15.2).

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4.15.3 Altitude. The equipment shall be subjected to test Method 500.1, Procedure I of MIL-STD-810 except that the chamber internal pressure of step 2 shall be 10,000 feet above sea level. The test procedure shall include a 17,500 feet altitude non-operational exposure for not less than 1 hour previous to the 10,000 feet altitude operational portion of step 2. The test item operation is to be verified during the elevation phase of step 2 at 10,000 feet above sea level in accordance with test requirements of 4.19 (See 3.15.3).

4.15.4 Immersion. The equipment shall be subjected to the test of Method 512.1, Procedure I of MIL-STD-810 (See 3.15.4).

4.15.5 Vibration.

4.15.5.1 Resonance search. The test item shall be secured to a vibration table that can be controlled within ±10 percent of the specified amplitude. Mounting method shall be such that the vibration within the test item can be observed and measured. To facilitate this observation and measurement, sub-assemblies may be tested separately, provided they are secured to the table in a manner similar to that used to mount them in the test item. The shock mounts (if any) of the test item shall be blocked. The test item shall be vibrated successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be 0.030 inch. In each of the three directions, the frequency shall be changed in steps of 1 cycle per second and maintained for at least 10 seconds. Vibration amplitudes shall be measured by optical or any other means, provided that the vibration of the test item is not affected by the measurement. The equipment shall have no resonance in the frequency range of 10 to 55 cycles per

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second that exceed twice the amplitude of the applied vibration, except that in the event that the keyboard fails to comply with the twice amplitude resonance limit, the keyboard, mounted on the equipment, shall be subjected to the resonance dwell. The dwell time shall be 15 minutes at peak resonant frequency point found to exceed 2-1/2 times the input amplitude. The input shall be a vibratory displacement of 0.030 inches. At the conclusion of the dwell period there shall be no degradation in performance or evidence of physical failure (seee 3.15.5.1).

4.15.5.2 Bounce, vehicular. The equipment shall be subjected to the test of Method 514.2, Procedure IX, Part 2 (equipment operating), of MIL-STD-810. Error rate may be degraded to not more than one error in processing 20,000 characters averaged over the test with a maximum of three errors per page. The equipment will be operated at a speed of 150 bps in the ASCII code for the duration of the test. (Operating bounce only). (See 3.15.5.2).

4.15.5.3 Bounce, loose cargo. The equipment in its field transport mode complete with paper supply roll and cover-stored items (copy holder and AC power cable) shall be subjected to the test of Method 514.2, Procedure XI, Part 2 of MIL-STD-810. After each 1/2 hour cycle, the verification test in accordance with 4.19 shall be performed (See 3.15.5.3).

4.15.6 Shock. Proper operation of each equipment after each shock test shall be checked by performing the verification test in accordance with 4.19 (See 3.15.6).

4.15.6.1 Drop. The equipment shall be subjected to the test of Method 516.2, Procedure II of MIL-STD-810. Minor damage (see 6.4.1) is permitted (See 3.15.6.1).

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4.15.6.2 Bench handling. The equipment shall be subjected to the test of Method 516.2, Procedure V of MIL-STD-810 (See 3.15.6.2).

4.15.7 Ballistic shock. (See 3.17.7)

a. Use the shock testing machine for lightweight equipment as shown in Figure 1 of MIL-S-901.

b. Mount the unit under test on a shock mount fabricated in accordance with CECOM drawing SC-D-972467. Fasten the shock mounted unit under test to the test plate in its normal operating position using the same fasteners used for vehicular installation.

c. Establish compliance with the operational requirements under standard ambient environmental conditions.

d. Maintenance of software integrity shall be verified for each blow via the following procedure:

(1) Set the unit under test to the ICT operational mode and apply power.

(2) Prior to each blow, fill the message memory of the unit under test via message reception to cause automatic printing of the received message in accordance with paragraph 3.8.7.1.1.2.

(3) With automatic printing of the received message in progress, subject the unit under test to the shock blow.

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(4) The unit under test shall accurately print the received message both before and after each blow. Garbling of the printed characters during and up to 1 second after impact is permissible.

e. Subject the unit under test as mounted and in accordance with the operational requirements of paragraph 4.15.7d above to a total of nine blows, one each of the following: 1 foot blow, 3 foot blow and 5 foot blow on the back, side and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible. Minor damage (See 6.4.1) is permitted.

f. Upon completion of the test, the verification test in accordance with 4.19 shall be performed.

4.15.8 Dust. The equipment shall be subject to the test of Method 510.1, Procedure I of MIL-STD-810. The equipment shall be in operating configuration during this test. After the dust test the verification test in accordance with 4.19 shall be performed. Penetration of dust into the equipment without causing an operational problem shall not be regarded as a failure (See 3.15.8).

4.15.9 Rain. The equipment shall be subjected to the test of method 506.1, Procedure I of MIL-STD-810. The equipment shall be powered on during the rain test, however, printing/feeding of paper shall not be required during rainfall. The paper shall be torn off against the tear bar after initialization and will remain in this position during the rain test. After the rain test, the verification test in accordance with 4.19 shall be performed. Penetration of rain into the equipment without causing an operational problem shall not be regarded as a failure (See 3.15.9).

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4.15.10 Salt fog. The equipment shall be subjected to the salt fog test specified in Method 509.1, Procedure I of MIL-STD-810. No corrosion products shall be deposited on the test item by the facility. After the salt fog test, the verification test in accordance with 4.19 shall be performed (See 3.15.10).

4.15.11 Orientation. The equipment shall be inclined for a minimum of 5 minutes in each plane (forward, backward, left and right) to an angle of 20 degrees, plus 3, minus 1 degree. During inclination in each plane, the equipment shall be operated and the verification test in accordance with 4.19 shall be performed (See 3.15.11).

4.15.12 Fungus. The equipment model in the assembled and as ready for delivery condition, or representative samples subject to approval of the procuring agency, shall be subjected to test Method 508.1, Procedure I of MIL-STD-810 except that the temperature humidity conditions of step 3.1.6a shall be changed to a constant 29 degrees C (84 degrees F) and 95% RH (minimum) during the life of the test. The equipment test model or samples shall not be specially cleaned for the fungus test, except for the cleaning it receives during or after production. All enclosed or gasketed assemblies shall be opened and the interior sprayed with the specified mixed spore suspension. After 28 days, based on visual examination, the equipment assembly or samples shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10% or less of the total area and no more than six unrelated minute colonies with mycelial development in areas only in other than critical circuit portions; such as terminal spacing, printed circuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. The equipment shall fail if it shows more than the growth specified above.

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4.15.13 Air seal test. The equipment shall be opened and closed again in such a manner as to break and remake the seal. Immediately thereafter the equipment, as field transported, shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the combination case. The vacuum then shall be valved off and the interior pressure measured during the ensuing period of one minute. During this one minute period the decrease in the vacuum shall not exceed 0.01 pound. The gauge used for measurement of this vacuum shall be of such accuracy that a difference of 0.01 pound can be determined accurately (See 3.15.13).

4.16 Reliability testing.

4.16.1 Test level. All reliability testing shall be conducted under the environment of Category 1, 50.1 (Appendix B) of MIL-STD-781C except that the keyboard shall be tested at ambient level room temperature during its operational cycle. During test, all equipment under test will be mounted without shock mounts in the normal attitude. Ten minutes out of every hour, the equipment shall be vibrated vertically with a sinusoidal motion having a frequency between 20 and 60 Hz. Throughout the test, the ambient will be maintained at 122 degrees plus nine degrees minus zero degrees Fahrenheit. Humidity will not be controlled except that it will not be allowed to go below 5 percent. The chamber temperature and equipment vibration shall be monitored and recorded in the test log.

4.16.2 Duty cycle. Each equipment shall have a 100% duty cycle. The keyboard shall have 50 percent duty cycle operating at a 50 words-per-minute typing rate. Keyboards shall be alternated on a once a week basis between the exerciser and operating modes.

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4.16.2.1 Exerciser. The keyboard shall be mounted at room temperature on a keyboard exerciser and each of the keys shall be depressed sequentially at a five character per second rate. All operating input voltages shall be applied to each keyboard while it is being exercised. The keyboards shall continue to be exercised for 168 hours.

4.16.2.2 Operating mode. Each keyboard shall be mounted on its printer in the normal operating mode. Each equipment shall continue being subjected to the vibration and temperature environments described previously. After 24 hours of testing, each keyswitch shall be operated sequentially 5 times (in the temperature environment) with the printer monitoring the keyboard. The keyboards shall continue to be operated for 168 hours.

4.16.3 Speed, code and interface. During first article and production reliability testing, two samples shall operate at 75 baud, 7 unit Baudot code, with a high-level interface, and the remaining samples at the highest speed, 10 unit ASCII code with a low-level interface.

4.16.4 Input voltage. Operating time of each system shall be equally divided between 24 Vdc and 115V, 60 Hz with a tolerance of +5%, -2%. Half of the systems shall be operated in each condition and alternated on a once a week basis.

4.16.5 Preventive maintenance. The equipment shall be capable of operation for at least 500 hours between lubrication periods or before adjustments are required to compensate for degradation of performance.

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4.16.6 Operator maintenance. Operator maintenance during this test shall consist of replacing the ribbon on the printer and replacing paper on the printer. All operator maintnenance will be recorded in the test log.

4.16.7 Test time. The minimum test time per system shall be in accordance with MIL-STD-781C, Test Plan VIIC . For reliability testing, test time shall include only time when the systems are operating. Off or stand-by time shall not be included in the calculation of the system's reliability.

4.16.8 Sample selecton. The equipment used for reliability testing shall be randomly selected by an authorized Government representative.

4.16.9 Classification of events.

4.16.9.1 Failure. Any electrical, mechanical, or adjustment malfunction which causes the equipments to print, transmit or receive improperly. Failure will be classified in accordance with paragraph 5.6.1 of MIL-STD-781C.

4.16.9.1.1 Non-relevant failures. The following failures will be classified as non-relevant:

a. Failures caused by operator maintenance.

b. Failures caused by operator error.

c. Normal replacement of consumable supplies of paper, and ribbons.

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d. Failures of test equipment or facilities.

e. Failures caused by operation of the equipment outside of its specified design.

f. Failures caused by act of God.

4.16.9.1.2 Relevant failures. All failures except those listed above are relevant failures unless determined otherwise by the procuring activity.

4.16.10 Errors. An error is an unrelated non-repeated failure of undetermined cause.

4.16.10.1 Error rate. The AN/UGC-74(A)(V) (with keyboard) shall not make more than one error in processing one million characters averaged over the test.

4.16.10.2 Chain of errors. In the system test to be conducted, an error caused by a unit will cause errors to appear in the other units of the system. Because the system is closed looped these errors will repeat until the error has been eliminated. Only the initial error of the system errors will be counted as one error.

4.16.11 Failure actions. Paragraph 5.6.2 of MIL-STD-781C shall apply. Entries shall be made on the data sheets for each failure and failed equipment shall be removed from test and repaired. When an intermittent failure occurs the equipment may be allowed to continue on test until the frequency of the occurrence becomes great enough to allow determination of the cause. Until a repair is made, these occurrences will be considered as one failure unless an unrelated new symptom occurs which may then be considered an additional failure.

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Verification of the repair may be accomplished by operating the equipment up to four hours only if specified in advance (in the test environment, if necessary). This verification time shall not be included as part of the statistical "on time". Failures that occur during this time will not be counted as part of the accept-reject criteria even if they are not related to the initial failure. They will, however, be recorded and reported to the procuring agency. Determination of compliance with the accept/reject criteria shall be in accordance with paragraph 5.7 inclusive of MIL-STD-781C.

4.16.12 Conditions precedent to reliability testing. Prior to performing any reliability acceptance testing, the following actions shall be completed in order as follows:

4.16.12.1 Group A and B testing. Reliability samples shall be selected from lots which have passed group A and B testing.

4.16.13 Functions to be performed. During the reliability tests the transmit, receive and print functions shall be performed. Details of the configuration and monitoring required shall be specified in the contractor's reliability test procedure.

4.16.14 Post test verification. After completing each reliability test, all test samples shall be resubjected to the group A tests. The results of group A testing shall be recorded and shall become part of the reliability test.

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4.16.15 Sample log forms. Test records and logs shall be similar to and contain all information as the forms in Figures 3 through 8 of MIL-STD-781B.

4.16.16 Test results. The results of the reliability demonstration testing shall be analyzed as follows:

a. Analyzed in accordance with paragraph 5.13 of MIL-STD-781C.

b. Recommendations for improvement of equipment system reliability.

4.17 Acoustics. Tests shall be conducted to measure compliance with the minimum noise levels (during operation) specified in MIL-STD-1474 (modified by table IV) (See 3.12).

4.18 Quality conformance inspection of packaging. Quality conformance inspection for packaging shall be in accordance with group A, group B and when required (see 6.1e), the rough handling test of MIL-P-116. Inspection lots shall be in accordance with MIL-P-116.

4.19 Verification test - Terminal, Communications AN/UGC-74A(V).

4.19.1 Purpose. The purpose of this test is to determine if any degradation of the equipment occurred during or after any environmental test to which the equipment was subjected.

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4.19.2 Equipment.

a. Teletypewriter under test. Set to ICT Mode.

b. One message generating device capable of transmitting a test message.

c. One message receiving device.

d. Equipment interconnection

(1) Low level

(2) ASCII (11 unit)

(3) 1200 bps

(4) Non-inverted data

(5) Odd Parity

(6) NRZ formatted data

4.19.3 Verification test.

a. Visually inspect equipment for damaged parts or assembly, corrosion and loose fasteners.

b. Turn power on (if necessary)

c. Perform SELF test

d. Receive test message

e. Print out received message

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f. Append message to header

g. Transmit

h. Verify correct reception of message

i. Remove messages from memory and verify

4.19.4 Acceptance. The criterion of successful completion is that the equipment passes each test above and meets the requirements of Section 3, paragraph 3.2.6, MIL-STD-810 upon completion of tests.

4.20 Maintainability demonstration. The maintainability requirements shall be demostrated in accordance with Notice 1 MIL-STD-471, Test Method 9. The sample size (n_c) of corrective maintenance tasks to be performed for the demonstration of the mean corrective maintenance time (u_c) shall be fifty (50) at each specified maintenance level. The specified mean corrective maintenance time (u_c) shall be 30 minutes at the organizational level and 2-1/2 hours at the direct support level.

a. The consumers risk (β) for demostration of uc shall be 10 percent. The equipment will be considered to have met the requirements when the specified parameters have received accept decisions. If the specified parameters fail to meet the accept decisions, then design corrective action shall be initiated at contractor's expense. When proper design corrective action has been completed, the equipment shall be retested at those levels of maintenance that previously failed. The same accept-reject criteria shall apply in the retest phase.

b. The contractor shall be responsible for data collection during the maintainability demonstration. This shall be accomplished by proper use of an accurate time recorder.

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The time shall be recorded on appropriate data sheets. These data sheets shall have provisions for the recording of the following times on each task to be demonstrated:

(1) Preparation

(2) Localization

(3) Isolation

(4) Disassembly

(5) Interchange

(6) Reassembly

(7) Alignment

(8) Verification checkout

c. While initiating corrective action during the demonstration, the maintenance technician shall follow procedures outlined in the equipment technical manual and none other, except where approved by the procuring activity.

d. The demostration test team shall consist of:

(1) A contractor director - introduces faults within the equipment.

(2) A contractor maintenance technician - performs maintenance tasks.

(3) A contractor data collector.

(4) Government witnesses.

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e. Maintenance tasks to be simulated shall be selected by Government representatives from a pool of proposed maintenance tasks submitted by the contractor at least two weeks prior to the actual demonstration. Task selection shall be in accordance with MIL-STD-471, Appendix A. Also, the pool of proposed maintenance tasks shall be subject to Government approval.

f. Fault simulation shall be accomplished in the following manner:

(1) Introducing an internally shorted or opened component.

(2) Introducing a similar component with a different value or rating.

(3) Introducing misalignment by detuning.

(4) Other contractor submitted and approved methods.

4.21 Weight. Weigh the equipment to verify conformance to 3.19.

4.22 Systems safety test. A visual and mechanical inspection shall be performed to verify conpliance with those portions of 3.17 which can be determined visually.

5. PACKAGING.

5.1 Packaging requirements. Packaging requirements shall be in accordance with SPI 1G00173.

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6. NOTES.

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specificatin and any amendment thereto.

b. Type required.

c. Date of DODISS (see 2.1.1).

d. The number of first article samples (see 4.3.1)..

e. When rough handling test is required (see section 4).

f. Technical literature required.

g. Quantity and kind of running spare parts required.

h. List of Cable assemblies to be Furnished by the Government.

i. Nuclear test calibration certification (see 4.14.1).

j. Level A or B preservation and packing (see section 5).

6.2 Cables. The following cables are used with, but not part of the AN/UGC-74A(V). If not furnished as GFP, the contractor shall fabricate cables as required for testing purposes:

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Procurement	Description	Number Reg.
SM-D-764480	Cable Assy, Electrical, DC	1
SM-D-764481	Cable Assy, Power, Electrical, 115 VAC	1
SM-D-764482	Cable Assy, Power, Electrical, 230 VAC	1
SM-D-915889	Cable Assy, XMIT/RCV High Level	1
SM-D-915896	Cable Assy, Lo Level Data, XMIT/RCV	1
SM-D-915897	Cable Assy, Lo Level XMIT/RCV Clock/Data	1
SM-D-915890	Cable Assy, Battery Back-Up	1

6.3 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited to severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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6.4 Definitions.

6.4.1 Minor damage. Minor damage is defined as chips and scratches of the paint finish and dents in the combination case that do not interfere with:

a. Sliding the equipment chassis in and out of its combination case to replenish tape or paper.

b. Compliance with the requirements of the Verification Test, Paragraph 4.19.

c. Compliance with the requirements of the Immersion Test, Paragraph 4.15.4.

6.4.2 Characteristic distortion. Distortion caused by transient or ripple voltage deviations occurring after some specified unit pulse interval. Only deviations occurring after 35% of the space signal time interval that deviate more than +0.15 or -.075 V from the nominal +1.5 V space signal are to be considered as not meeting requirements (NIK only).

6.4.3 Error. An error is an unrelated, non-repeated event of undetermined cause.

6.4.4 Failure. A failure is defined as any electrical, mechanical or adjustment malfunction which causes the equipment to operate improperly.

6.4.5 Performance requirements of the equipment. Whenever referenced in this specification, the "performance requirements of the equipment" is to be understood as a reference to all the electrical and mechanical performance specified and implied by this equipment specification.

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6.4.6 Module. The smallest subassembly that is replaced during a repair activity. A module is an integrated set of components (see 6.4.12) that is removed and replaced as a whole. An example of a module is a printed wiring card and its electronic components. A module may be either repairable or non-repairable.

6.4.7 Component. The smallest part that is replaced during a repair activity. All components are considered to be non-repairable throwaway items.

6.5 Nomenclature. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.6 Color. The color chip furnished by the procuring agency will be lusterless forest green (Film V or W), and may be obtained upon request to the Contracting Officer.

6.7 Government furnished equipment (GFE).

6.7.1 System interface test. The following GFE will be provided for the system interface test of 4.10.6; KW-7 (with both BAUDOT and ASCII capability).

6.8 System configurations. The AN/UGC-74A(A) communications terminal configurations are shown in table XI.

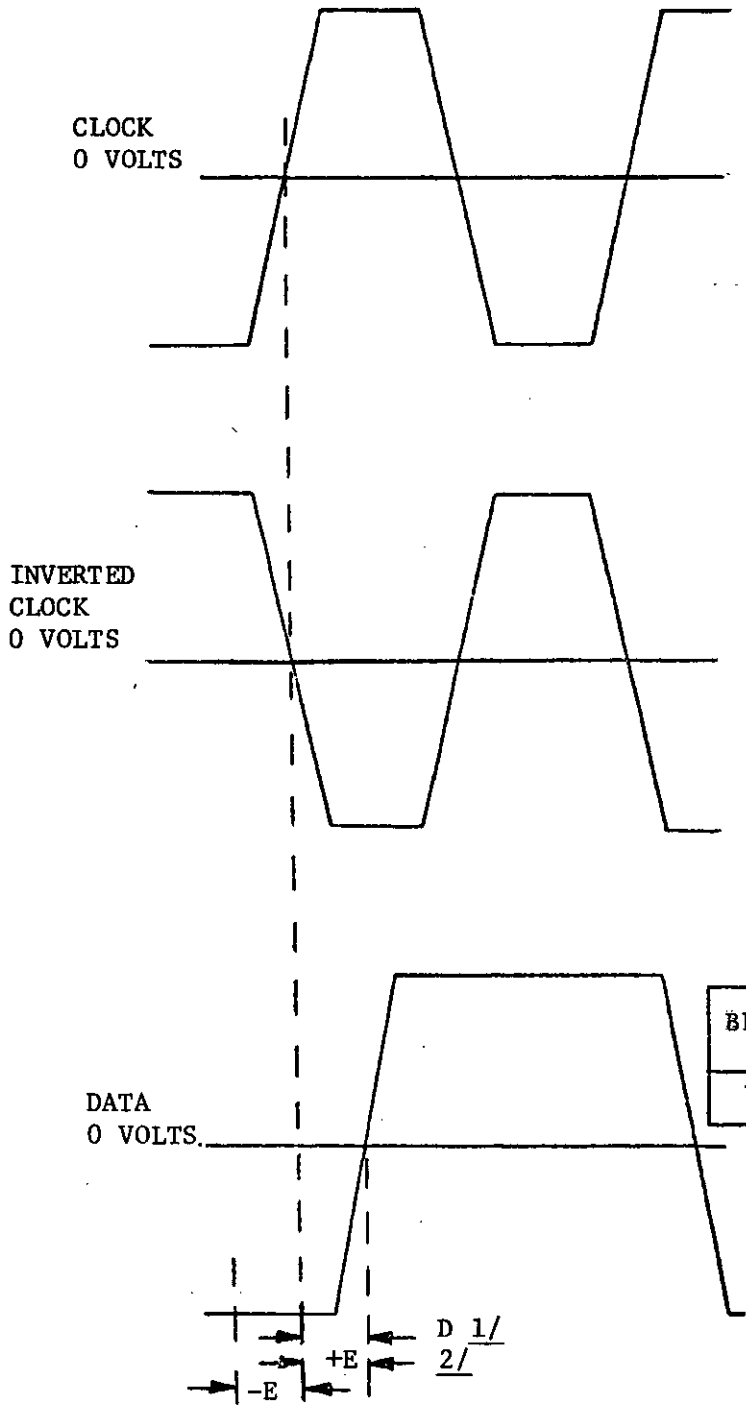
6.9 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

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TABLE XI. Terminal, Communication AN/UGC-74A(V) configurations.

ITEM	NOMENCLATURE	TELEPRINTER TT-744/ UGC-74A(V)	KEYBOARD TT-745/ UGC-74A(V)	CASE, TELEPRINTER CY-7735/ UGC-74A(V)	COVER, TELEPRINTER MX-10038/ UGC-74A(V)
RECEIVE-ONLY (RO) PAGEPRINTER	TERMINAL, TELEPRINTER AN/UGC-74A(V)1	X		X	X
	TERMINAL, TELEPRINTER AN/UGC-74A(V)2	X		X	
KEYBOARD SEND-RECEIVE (KSR) PAGEPRINTER	TERMINAL, COMMUNICATION AN/UGC-74A(V)3	X	X	X	X
	TERMINAL, COMMUNICATION AN/UGC-74A(V)4	X	X	X	

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BPS	E=4.5% TR CLOCK	D=12% REC CLOCK
75	± .6	1.60

1/ D is a maximum of 12% (REC CLOCK) of bit length

2/ E is a maximum of ± 4.5% (XMIT CLOCK) of bit length

FIGURE 1. Relationship of transmitted data to clock.

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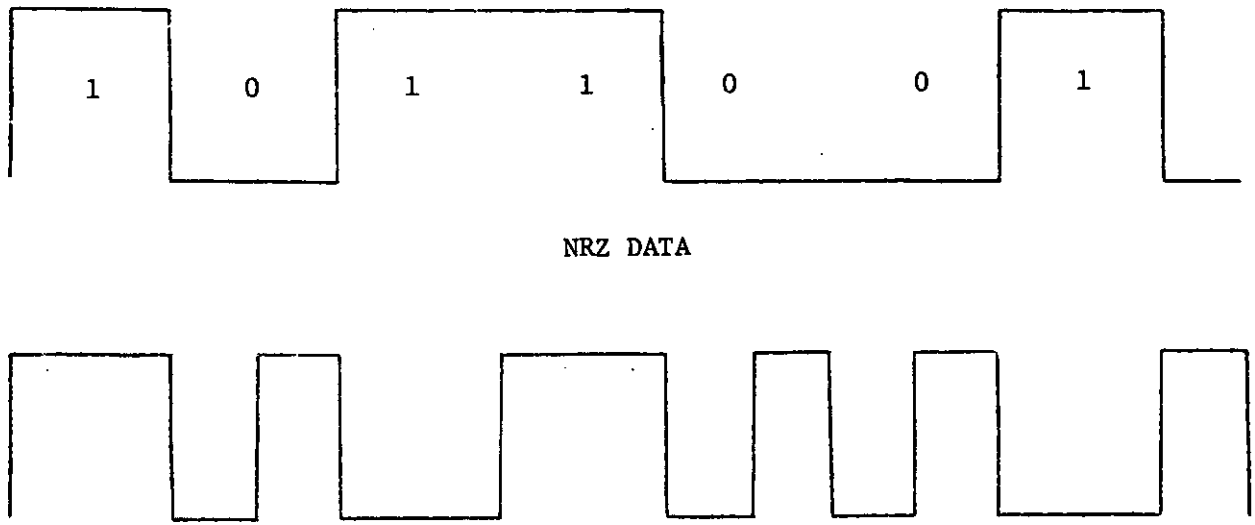


FIGURE 2. Conditioned diphase modulated output

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